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Contextual Report

AquaIntel: A Domain - Specific AI System for Smart Aquarium Management

Submitted To:

Supervisor:

Ajaya Kumar Sharma

Project Coordinator:

Pawan KC

Submitted By:

Amogh Shakya

2337878

ABSTRACT

AquaIntel is a mobile-based domain-specific AI system developed to enhance and simplify aquarium management through intelligent automation and focused decision support. Traditional aquarium maintenance often relies on manual monitoring and general knowledge, which can lead to inconsistent care and potential risks to aquatic life. AquaIntel addresses these challenges by integrating a domain-restricted AI chatbot that provides accurate, aquarium-specific guidance while filtering out unrelated information.

The application features a smart reminder system that notifies users about essential maintenance tasks such as fish feeding, water changes, and tank cleaning, ensuring a consistent care routine. Additionally, an interactive dashboard enables users to monitor key aquarium parameters, including pH levels and overall tank conditions, promoting proactive management.

By combining artificial intelligence, mobile accessibility, and real-time monitoring, AquaIntel offers a user-friendly and efficient solution for aquarium owners. The system aims to improve fish health, reduce manual effort, and provide reliable, data-driven insights, making aquarium management more accessible for both beginners and experienced users.

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Name: Amogh Shakya

University ID: 2337878

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1. Introduction

Aquarium management is a delicate and continuous process that requires careful monitoring of water conditions, regular maintenance, and proper fish care practices. Maintaining factors such as pH levels, temperature, and cleanliness is essential for ensuring a healthy aquatic environment. However, many aquarium owners, especially beginners, often rely on manual methods and general online information, which can lead to inconsistent care and potential harm to aquatic life.

With the rapid advancement of mobile technology and artificial intelligence, there is an opportunity to improve traditional aquarium management through smart, automated solutions. In particular, domain-specific AI systems have emerged as a powerful approach for delivering accurate and context-aware information within a focused area of knowledge. Unlike general-purpose AI, domain-restricted systems can provide more reliable and relevant responses by limiting their scope to a specific subject.

This project introduces AquaIntel, a mobile-based domain-specific AI system designed to assist users in managing aquariums more effectively. The application incorporates a domain-restricted AI chatbot that responds exclusively to aquarium-related queries, ensuring precise and trustworthy guidance. In addition, AquaIntel includes features such as maintenance reminders, water condition monitoring (including pH levels), and an interactive dashboard for tracking aquarium health and tasks.

The primary aim of AquaIntel is to simplify aquarium care by combining intelligent assistance with user-friendly mobile functionality. By reducing reliance on manual processes and providing timely insights, the system seeks to enhance fish well-being, improve maintenance consistency, and support users in making informed decisions regarding their aquariums.

1.1. Background

Aquarium keeping has become a popular hobby and practice worldwide, valued for its aesthetic appeal and the calming effect it brings to indoor environments. However, maintaining a healthy aquarium ecosystem is a complex task that requires consistent attention to multiple factors, including water quality, feeding schedules, filtration, and fish compatibility. Even small imbalances in parameters such as pH levels, temperature, or ammonia concentration can negatively affect fish health and may lead to disease or mortality.

Traditionally, aquarium management relies heavily on manual monitoring and the personal experience of the aquarium owner. Many users depend on generic online resources, forums, or trial-and-error approaches to manage their tanks. While such methods can be helpful, they often lack personalization and may provide inconsistent or inaccurate information, especially for beginners who have limited knowledge of aquatic ecosystems.

In recent years, advancements in mobile applications, Internet of Things (IoT), and Artificial Intelligence (AI) have opened new possibilities for improving aquarium management. Smart aquarium systems now allow users to monitor environmental conditions through sensors, automate routine tasks, and receive alerts when parameters go beyond safe limits. Additionally, AI-driven solutions have shown potential in providing intelligent recommendations, predictive insights, and real-time decision support.

Despite these developments, many existing systems are either expensive, overly complex, or rely on general-purpose AI models that are not specifically tailored to aquarium-related knowledge. This creates a gap for a more focused and accessible solution that combines intelligent assistance with ease of use.

AquaIntel is developed in response to this need, aiming to leverage a domain-specific AI approach to provide accurate, relevant, and user-friendly support for aquarium management. By integrating features such as a restricted AI chatbot, maintenance reminders, and an interactive monitoring dashboard into a mobile platform, AquaIntel seeks to bridge the gap between traditional aquarium care and modern intelligent systems.

1.2.Problem Statement

Aquarium management requires continuous monitoring and proper maintenance to ensure a stable and healthy environment for aquatic life. Key factors such as water quality, feeding schedules, and tank cleanliness must be carefully maintained; however, many aquarium owners struggle to manage these effectively. This is especially challenging for beginners who lack sufficient knowledge and rely on inconsistent or generalized online information.

Existing solutions for aquarium care are often limited in functionality. Traditional methods depend heavily on manual tracking, which is time-consuming and prone to human error. While some digital tools and smart systems are available, they may be expensive, complex, or not specifically designed for aquarium-related needs. Additionally, general-purpose AI chatbots can provide inaccurate or irrelevant responses due to their broad knowledge scope, making them less reliable for domain-specific guidance.

As a result, users often face issues such as irregular maintenance, poor water quality management, missed feeding schedules, and lack of timely alerts. These problems can negatively impact fish health and lead to avoidable losses.

Therefore, there is a need for a mobile-based, user-friendly system that integrates domain-specific AI, automated reminders, and real-time monitoring to provide accurate guidance and assist users in maintaining their aquariums efficiently. AquaIntel aims to address these challenges by offering a focused and intelligent solution tailored specifically for aquarium management.

1.3.Proposed Solution

To address the challenges of traditional aquarium management, this project proposes AquaIntel, a mobile-based domain-specific AI system designed to provide intelligent, accurate, and user-friendly support for aquarium owners.

The core of the solution is a domain-restricted AI chatbot that is specifically designed to handle only aquarium-related queries. Unlike general AI systems, this chatbot filters out unrelated topics and delivers precise information about fish care, water conditions, feeding practices, and maintenance. This ensures reliability and reduces the risk of misinformation.

In addition, AquaIntel introduces a smart reminder and notification system that helps users maintain a consistent care routine. The system sends automated alerts for essential tasks such as:

- Fish feeding schedules
- Water changes
- Tank cleaning and maintenance

This feature minimizes the chances of missed activities and supports better aquarium health.

The application also includes an interactive dashboard that allows users to monitor and manage their aquarium efficiently. Users can track important parameters such as pH levels and maintain records of tank conditions over time. The dashboard presents data in a clear and visual format, enabling users to quickly understand the state of their aquarium and take necessary actions.

Furthermore, AquaIntel is designed as a mobile application, making it easily accessible and convenient for users to manage their aquariums anytime and anywhere. The system focuses on simplicity, ensuring that both beginners and experienced users can use it without difficulty.

Overall, the proposed solution combines artificial intelligence, automation, and real-time monitoring to create a smart aquarium management system that improves fish well-being, reduces manual effort, and provides accurate, domain-specific guidance.

1.4.Aim and Objectives

The main aim of AquaIntel is to develop a domain-specific AI system that provides accurate, reliable, and safe expert guidance for aquarium care, minimizing misinformation and improving the health and sustainability of aquatic ecosystems.

Objective

- * To design and develop a specialized AI chatbot focused exclusively on aquarium care and aquatic life management.
- * To implement a secure and scalable backend system using ASP.NET Core and MongoDB for efficient data processing and storage.
- * To create a structured aquatic species database that supports diverse species attributes and flexible data modeling.
- * To ensure high reliability of AI responses by restricting knowledge boundaries to domain-specific information.
- * To enhance user experience through intuitive interaction, quick response times, and accurate expert-level recommendations.
- * To demonstrate practical application of controlled AI systems suitable for real world deployment in hobbyist and professional environments.

1.5.Scope

The scope of the AquaIntel project is focused on developing a mobile-based application that assists users in managing aquarium environments through domain-specific AI and smart monitoring features. The system is designed to support individual aquarium owners by providing tools for maintaining fish health and tank conditions efficiently.

AquaIntel includes a domain-restricted AI chatbot that responds only to aquarium-related queries, ensuring accurate and relevant information. The application also provides a reminder and notification system to help users manage routine tasks such as feeding, water changes, and tank maintenance.

The system offers an interactive dashboard where users can input and monitor key aquarium parameters, such as pH levels and general tank conditions. It focuses on providing basic monitoring and guidance rather than full automation through hardware devices. The application is intended to be user-friendly and accessible, making it suitable for both beginners and experienced aquarium enthusiasts.

However, the scope of the project is limited in certain areas. AquaIntel does not include direct integration with physical IoT sensors or automated hardware control in its current version. The AI chatbot is restricted to predefined aquarium-related knowledge and may not cover highly advanced or rare cases. Additionally, the system relies on user input for data such as water parameters, which may affect accuracy.

Overall, the project is scoped as a software-based intelligent assistant for aquarium management, with potential for future expansion into IoT integration and advanced AI capabilities.

1.6.Intellectual Challenge

Developing AquaIntel posed several intellectual challenges that required deep problem-solving and creative thinking:

Domain-Specific AI Constraints – Designing an AI chatbot restricted to aquarium-related queries only was challenging. Ensuring the AI could understand varied user questions while avoiding irrelevant topics required careful prompt design and filtering logic.

Intelligent Notification System – Implementing reminders for feeding, water changes, and maintenance involved creating a reliable scheduling system that could handle multiple fish tanks and different species’ needs.

Data Handling and User Interaction – Structuring and storing aquarium data efficiently while allowing the AI to access it contextually demanded thoughtful database design and integration.

User Experience and Accessibility – Designing an intuitive mobile interface that lets users interact easily with AI while receiving alerts and maintaining their aquarium required balancing simplicity with functionality.

Integration of Multiple Functionalities – Combining AI conversation, reminders, alerts, and user data management into a single cohesive mobile application tested both logical planning and coding skills.

These challenges strengthened problem-solving, algorithmic thinking, and practical implementation skills, which were essential to building a functional and intelligent aquarium assistant.

1.7. Rationale

These challenges strengthened problem-solving, algorithmic thinking, and practical implementation skills, which were essential to building a functional and intelligent aquarium assistant.

Maintaining an aquarium can be complex and time-consuming, especially for beginners or enthusiasts managing multiple fish species. Fish require proper feeding schedules, clean water, and suitable environmental conditions, and missing these can lead to poor fish health or loss. Traditional methods rely on manual tracking, which is prone to errors and forgetfulness.

AquaIntel aims to address these challenges by providing a smart, domain-specific AI assistant that helps users manage their aquariums efficiently. By offering contextual advice, reminders, and alerts tailored to specific fish species and tank conditions, the application reduces human error, simplifies aquarium maintenance, and enhances the overall experience of keeping fish.

This project is motivated by the need for a reliable, intelligent, and user-friendly solution that combines AI capabilities with practical aquarium management tools.

1.8. Contribution

The AquaIntel project contributes to the field of smart aquarium management in the following ways:

Domain-Specific AI Assistance – Provides an intelligent chatbot that answers only aquarium-related queries, offering accurate and context-aware guidance to users.

Automated Reminders and Notifications – Helps users maintain their aquariums efficiently through alerts for feeding schedules, water changes, and other maintenance tasks.

Enhanced User Experience – Simplifies aquarium management by integrating multiple functionalities into a single mobile application, reducing errors and effort for both beginners and experienced fish keepers.

Educational Value – Offers informative advice about fish care, tank management, and environmental factors, promoting better understanding and responsible fish keeping.

Innovation in Practical AI Use – Demonstrates the application of AI in a focused domain, showing how technology can solve specific real-world problems in a targeted, user-friendly way.

1.9. Thesis Structure for AquaIntel

Chapter Name	Contents
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Critical analysis/evaluation	6.1 Testing Strategy/ plan 6.2 Evaluation Strategy /plan

Figure 1 Thesis Structure

2. Literature Review

Literature Review Report: AquaIntel System

Theme 1: AI Algorithms & Compatibility Logic

2.1. Random Forest Classification for Aquatic Compatibility (Khan, et al., 2025)

- **What does it say?** Khan established that predictive modeling using Random Forest algorithms can determine species compatibility with over 90% accuracy by analyzing multivariate data (pH, temperature, and aggression levels).
- **Implementation:** One of the most important lessons that we have learned in the work of Khan is the application of the ranking of Features Importance. The research points out that this algorithm works best when variables are weighted based on biological influence. An example is that temperament (aggression) and minimum tank size are much more important in the compatibility score than taxonomic genus or physical coloration. This makes sure that the model focuses on life-safety measures and not the aesthetic similarities.
- **Relevance:** The AquaIntel Compatibility Checker is based on this research as its mathematical basis. The application does not simply give a simple Yes/No answer but it combines the weighted logic of Khan to come up with Compatibility Percentage. This enables the user to be able to view the reasons behind some fish being high-risk (e.g., "95% water match, but only 20% behavioral match").
- **Improvement:** Although the model by Khan is sound, it mainly concentrates on the introduction of species at the beginning. To improve this in the case of AquaIntel, I will add logic of Maximum Adult Size. A large number of hobbyists will buy juvenile fish that are compatible in the short term, but become territorial or overgrow their habitat as they grow. With the introduction of a variable of growth-rate to the framework by Khan, AquaIntel will offer a more sustainable and long term projection of the ecosystem.

2.2. Weighted Compatibility Matrices & Swimming Strata (Revankar, 2025)

- **What does it say?** A 2025 study by Revankar suggests that aquarium failure is frequently due to spatial competition as opposed to water chemistry. The experiment determined that there is a compatibility spectrum, which is given by three essential dimensions: Water Chemistry, Temperament, and Swimming Strata (explicit vertical areas in which a fish will

be found, i.e., top, middle, or bottom). The information provided by Revankar indicates that even species with moderate aggressiveness can live side by side as far as they are in different strata since they seldom interfere with each other to fight over the territory.

- **Implementation:** The proposed system uses a relational database in which each species will be given a Strata ID. The compatibility score in the model by Revankar is dynamic: in case the system observes two semi-aggressive species sharing the same Strata ID (e.g. both are bottom-dwelling, territorial cichlids), the compatibility percentage decreases considerably. On the other hand, it determines areas of the tank that are vacant, and the system is able to recommend vacant areas to be occupied by Dither Fish which are active and peaceful fish that swim through open water to minimize the shyness and aggressiveness of other residents.
- **Relevance:** The AquaIntel Suggestion Engine is directly informed by this research. The app can now recommend fish that will balance the tank, as opposed to merely asking whether a fish can survive in the tank. When a user of the dashboard only sees bottom-dwelling species then AquaIntel will suggest the top-dwelling species to bring about a more aesthetically and biologically balanced setting
- **Improvement:** One of the weaknesses of the Revankar study is its assumption on the expertise of the user when it comes to fish behavior. To enhance this with the AquaIntel UI/UX, I will avoid technical terminologies such as Strata ID. Rather, I will apply visual signs to the interactive dashboard. The visual representation of the aquarium by displaying a simple tank graphic with icons in the top, middle, or bottom parts of the tank allows users to quickly see the spatial distribution of their aquarium without having to comprehend the underlying behavioral ecology

2.3. Predictive Disease Diagnosis via Symptom Logic (Shah, 2026)

- **What does it say?** In his study, Shah (2026) concentrates on the idea of detecting the earliest aquatic pathogens by behavioral and physical indicators. The research confirms that about 90 percent of typical aquarium illnesses, including Ich (Ichthyophthirius multifiliis), Velvet, and Dropsy, have apparent and predictable symptoms at their initial stage. Shah claims that with a developed system based on a systematic Symptom Logic

tree, an intelligent system could detect possible outbreaks prior to their terminal stage, and greatly enhance the survival rate of the occupants of a tank.

- **Implementation:** The author suggests a Symptom Checker interface, which allows users to provide observations through a standard list of physical and behavioral anomalies. To illustrate, when choosing white crystalline spots and clamped fins and flashing (rubbing against rocks), a high-probability diagnosis of Ich can be provided. This reasoning helps the AI to reduce complicated biological problems to actionable data without the user being a marine biology major.
- **Relevance:** This study gives the operational model of the AquaIntel Digital Vet feature. The AI assistant is capable of delivering real-time and step-by-step quarantine and triage information by incorporating the symptom logic of Shah. Once an user is deviated, the app can suggest to isolate the fish or change certain environmental conditions (such as increase the temperature or include salt in the aquarium) to stop the infection spread.
- **Improvement:** One of the ethical issues that arise strongly in Shah study is the possibility of users misunderstanding AI guidance as a formal veterinary prescription. To enhance this to the advantage of AquaIntel, I will introduce a "Safety-First" policy. The AI will be coded to offer a required medical disenfranchisement and will lean more towards a course of Natural Remedy - water changes and stress relief - before proposing chemical interventions. This is to make sure that the user is not likely to crash his or her biological filter with unwarranted medications.

2.4. The Nitrogen Cycle & Biological Rule-Sets (Sandford, 2023)

- **What does it say?** In any fish tank, the Nitrogen Cycle is the most significant process that Sandford describes in 2024. It demonstrates the transformation of fish waste into poison (Ammonia and Nitrite). For a tank to be a "balanced home," these poisons must be at zero. The study confirms that it requires approximately four weeks before the growth of good bacteria to be sufficient to purify the water organically. Fish cannot last long without this cycle.
- **Implementation:** In the study, there are Hardiness rating of various fish. This implies the arrangement of fish in order of toughness. Sandford suggests that "sensitive" fish should

never be put into a new tank until the 4-week cycle is finished. The first must be only hardy fish (those that can tolerate slightly dirty water) to assist the good bacteria begin growing.

- **Relevance:** This study provides the AquaIntel AI with its brain to be safe. It prevents the greatest mistake of beginners to spend too much money on purchasing expensive, fragile fish at too young age. When your tank is only a week old, the AI will inform you: "Your tank is not ready to receive this fish yet. It applies the rules given by Sandford to provide a security guard to your pets.
- **Improvement:** Most books describe the Nitrogen Cycle using the science words which are confusing to understand. I will not simply use text to make AquaIntel better. I will create a Visual Progress Tracker within the app. This will be a simple bar or graph which will indicate the user to the exact point of where his tank is to be safe. The app transforms waiting time (30 days) into a visual loading bar, which allows patients to wait and preserve their fish alive.

Theme 2: Notification Management (Combatting Fatigue)

2.5. Intelligent Alert Mechanisms & Cooldowns (Sarkar Akib, 2026)

- **What does it say?** A significant issue as outlined in a study by Akib (2026) is known as the Alarm Fatigue. This occurs when an application gives excessive notifications to the user, overwhelming them. The user eventually ceases to view the notifications or removes the app altogether. Akib discovered that repetitive or irrelevant pings will make the users lose their faith in the app as a useful tool.
- **Implementation:** The study recommends the application of "Configurable Cooldowns" to correct this. This implies that the app is coded politely. To say the least, in case the app reminds you to Clean the Tank Glass, but you do not do it immediately, the application will not remind you within at least 2 hours. It knows that you are in a hurry and it does not allow your phone to buzz after every five minutes to do a task, which is not an emergency..
- **Relevance:** This paper is the prototype of the AquaIntel Reminder System. It renders the app smarter and not as annoying. With the help of the logic of Akib, the system will divide Critical Alerts (such as Water Temperature is too high!), and Routine Tasks (such as Wipe

the lid). Immediate alerts are critical and routine tasks can be separated to prevent the user experience of being nagged.

- **Improvement:** I observed that Akib has conducted research on the issue of when to send an alert but not the content of the alert. Contextual Enrichment will be used to enhance this in the case of AquaIntel. Rather than firing off an automatic ping that merely states Tank Alert, the notification header will display the specific task like Change 20% Water Now. This makes the system highly efficient as the user is made aware of what they are supposed to do without even opening the app.

2.6. Intelligent Batching & Digest Logic (Courier Research, 2026)

- **What does it say?** Research from Courier (2026) shows that users are much more likely to keep an app if it doesn't interrupt their day constantly. They discovered that applying the concept of Time-based Batching, which involves combining multiple small alerts into one summary or Digest, can decrease app uninstalls by 65%. The user receives one structured update at a time instead of five separate pings throughout the day that would suit the user (such as 9:00 AM).
- **Implementation:** This study proposes a Value Hierarchy of all alerts. This implies that the app will have to rank all messages according to their importance. Low-Value: The daily summary postpones reminders such as "Check the light timer" until the daily summary. High-Value: Life-Threatening alerts only (such as a sudden fall in oxygen level or a broken heater) are permitted to bypass the filter and send an instant alert to the user of the phone.
- **Relevance:** This research is used to create the "Daily Tank Briefing" in AquaIntel. Rather than the app alerting me every time I need to complete such minor maintenance tasks, it will compile filters cleaning, trimming the plants, and wiping the glasses into a single morning notification. This will avoid the problem of Notification Fatigue and will help the user feel like the app is a helpful tool and not an unceasing disturbance.
- **Improvement:** Although batching is excellent when it comes to peace of mind, it may prove hazardous in case the user forgets about a task that is of high priority (such as a required water change). To enhance the research, I will put in place Nagging Intensity. After 48 hours neglecting a critical task, the app will raise the frequency of the alerts and

switch the notification color to red. This makes us certain that as we are being polite with petty chores that we never endanger the biological wellbeing of the tank by an inattentive user.

2.7. Severity-Based Routing for User Retention (Xurrent, 2026)

- **What does it say?** According to the study of Xurrent (2026), users can only trust an application when it understands that there is a minor update and a critical issue. When the same loud sound is used in the app to say Your tank looks nice today and to say Your fish are dying, the user will no longer take the app seriously. The experiment confirms that Severity-Based Routing, which routes various alerts along different routes, is both more engaging to users and helps avoid notification burnout.
- **Implementation:** To create this work, the system is coded to employ Silent vs. Active alerts. Critical Alerts: Only changes in status that are life-threatening (such as a 50% decrease in oxygen) enable a phone vibration or sound to be triggered. Routine Tasks: Daily (such as "It's time to have the monthly light cleaning) are dispatched as Silent notifications. They appear on the screen, yet do not make any sound to interrupt the life of the user. It educates the user that When the phone buzzes, it does matter.
- **Relevance:** This study determines the way in which the AquaIntel Reminder System communicates. It is so that the user does not feel overwhelmed. The app creates a professional connection with the hobbyist by dividing the concepts of Information and Action Required. The user is aware that whenever he or she hears a notification sound of AquaIntel, it means that he or she should check his or her tank.
- **Improvement:** Although Xurrent researches on the means of sending the alert, it does not address the issue of what to do afterwards. To enhance this in the case of AquaIntel, I will include "Self-Healing" suggestions. The notification will have a 1-click How-to-Fix guide instead of simply stating the message: pH is too low. This provides the user with an instant solution (such as "Add crushed coral" or "Perform a 10% water change) in order to be able to switch between Panic and Action in seconds.

2.8. Empowering Preferences in Mobile Notifications (IEEE, 2025)

- **What does it say?** One of the 2025 studies released by IEEE discusses how digital stress can be alleviated by providing users with more control over their notifications. The

researchers suggested a system that has three different modes: Immediate: Applied to life-saving notices which are urgent. While in Use: Information is only shown when the user actively opens the app. On-Demand: The user can check detailed statistics and reports on his/her own will. The experiment showed that such modes prevented users to habitually check their phones and therefore reduced their anxiety.

- **Implementation:** The study revealed that users become more empowered when they have the option of determining how fast they want the information to be delivered. In a fish tank scenario, this will mean that the app does not have to remind you every time a plant grows an inch; that can be stored until you are in the app. This makes it a lot less stressful to the hobbyist.
- **Relevance:** The study assists in determining the Notification Settings of AquaIntel. It will give the user an option on when he or she will be reminded of low-priority chores. As an example, a user may have the option of setting a reminder of "Filter Cleaning" to be shown only when the user is in Use, i.e. the reminder will only be displayed when the user opens the app to monitor the health of his/her tank. This will make sure that the app does not seem like it is nagging the user in the course of their work or family.
- **Improvement:** Although the IEEE study concentrates on the types of alerts, it does not focus on the right now aspect. To enhance this in the case of AquaIntel, I will include a Snooze button in the notification drawer. Whenever the user receives a reminder but is occupied, they do not need to go to the app or dismiss it, instead they can just tap Snooze 1 hour. This allows the user to have complete control over his or her schedule and make sure the task is not forgotten.

Theme 3: Offline Access & AI Performance

2.9. Local Database Caching for Zero-Latency (MDPI, 2025)

- **What does it say?** The MDPI journal (2025) research indicates that mobile applications do not work when they are 100% cloud-based. The paper demonstrates that there is a way to implement direct storage and management of SQLite databases on the phone using the term Service Workers that makes it possible to achieve a zero-latency. This implies that the app can be used immediately, even when there is a dead zone (no Wi-Fi or cell signal). The technology guarantees that the user does not even have to wait to get a loading screen when accessing valuable information.

- **Implementation:** According to the study, in the case of specialized apps, the whole core database (in your case, the list of fish species and their care needs) must be downloaded and stored locally when launching the app the first time. A species database is typically small (around 10MB), which means that it can be easily stored on any modern smartphone. This enables the app to do complicated "Compatibility Checks" all on the device without the need to communicate with a remote server.
- **Relevance:** This study is crucial to the AquaIntel Offline Mode. There are numerous pet stores in big metal constructions or cellars where 5G and Wi-Fi networks are sometimes unproductive or absent. Using the caching approach provided by MDPI, one can be at the center of a fish shop, enter the name of a species, and receive real-time AI suggestions despite the phone being in Airplane Mode.
- **Improvement:** One of the challenges that local databases are exposed to is that the information becomes obsolete. To enhance on this, I will use Background Sync. Rather than requiring the user to manually update the app, AquaIntel will automatically check whether there is new fish information or updated care instructions only when the phone is on unmetered Wi-Fi. This makes the pet store experience quick and offline-enabled and makes sure that the information is not older than few days.

2.10. On-Device Intelligence & ML Kit (Goodfirms, 2026)

- **What does it say?** The study is about the change of Cloud AI to On-Device Inference. It describes that contemporary smartphone chips contain special NPUs (Neural Processing Unit) that are specifically designed to perform AI operations. The AI can process data on the phone itself, instead of sending it to a remote server, which makes it much faster and more confidential since the personal tank data of the user will never need to leave the phone.
- **Implementation:** The study proposes to use Model Quantization to make this work on a mobile device. It is a technical procedure that reduces a huge AI model to a significantly smaller one. The model allows the phone to execute the math locally on the CPU and NPU because it is smaller than the AI. This will guarantee that the user receives a response within milliseconds, as opposed to having to round-trip to the cloud.

- **Relevance:** This is the "engine" that powers the AquaIntel AI Assistant. Since the AI is stored on the phone, it can perform optimally when there is no network connection, such as a cellar fish shop or a rural residence. It also makes sure that the chatbot is constantly ready to provide guidance on the compatibility of fish or water chemistry, no matter whether the user has an internet connection.
- **Improvement:** A typical threat of on-device AI is the fact that it is potentially intimidating, making the phone overheat and the battery run out fast. In order to enhance the simple research, I will apply principles of TinyML (Tiny Machine Learning). I will not employ a large general-purpose model, but a very specialized, small model that has heard of aquariums only. This helps to avoid heating up the device with the app and makes sure that AquaIntel will not be too heavy and battery unfriendly to use in daily life.

2.11. Dual-Mode Thinking for Mobile AI (SiliconFlow, 2026)

- **What does it say?** A recent study by SiliconFlow (2026) brings out the notion of Dual-Mode Thinking to mobile Artificial Intelligence. It has been demonstrated that an AI can be divided into two layers: a high-speed Chat Mode that allows an AI to engage in simple and everyday dialogues and a deep Thinking Mode that lets an AI solve complex problems. With an alternation between these two modes, the AI is able to give deep reasoning only where it is required, as opposed to squandering computer power on trivial questions.
- **Implementation:** This research determined that the basic interaction mode of the AI (Chat Mode) consumes 80 per cent less battery than the deep reasoning mode. This is a game-changer to a mobile app. It enables the software to remain quick and receptive to 90% of user inquiries and delegate the Heavy Thinking to hard tasks that need multiple data points to be analyzed simultaneously.
- **Relevance:** This powers the AquaIntel AI-Based Fish Assistant. When a user poses a simple query such as What do Guppies eat? then the app will remain in its low-power Chat Mode where the user gets a quick response. Nevertheless, in case the user tells about a complicated health crisis related to water chemistry and several symptoms, then the app activates Thinking Mode to cross-reference your database and offer a detailed and logical solution.

- **Improvement:** One of the biggest risks of mobile AI is the so-called hallucination whereby the AI fabricates facts that do not hold. This creates a threat to fish keeping. To enhance the work, I will Ground the AI in your local, verified database. This translates to the fact that, prior to the AI speaking, it has to check its response with the species facts in the phone. The AI cannot guess; when the information is not available in the proven local data, it should first make sure it is safe before providing a creative solution.

2.12. UX Design for "Zero-Internet" Reliability (Goodfirms, 2026)

- **What does it say?** This research identifies that apps lacking "Offline Indicators" which creates a bad impression of the app that can lead to a frustration to the user when the user cannot upload data in the app. This study simply helps to prevent the frustration of user by just adding offline indicator an clear message in the app so that user is informed and the app utilization can smoothly use the app.
- **Implementation:** This research implies using a optimistic UI which is a design pattern. The is acts as name says optimistic design even without internet it shows the cache data rather than a loading being displayed in the screen. This helps the app feel more responsive, fast, responsive even without the connection.
- **Relevance:** The AquaIntel Compatibility Checker is critical. When a user is in a dead zone (like a pet store basement) the application will still work correctly. However, in accordance with Vercel's recommendation, the application will show the "Last Updated" Timestamp. Meaning that the User is seeing data that was saved 3 hours prior. This transparency of what is being viewed as stored and still good data, as opposed to a broken or empty screen, is important for the user to understand.
- **Improvement:** With optimistic UI, the user may attempt to do something major, such as add 5 new fish into their digital aquarium, while they are offline. Once they go back online, they may forget that no information has been sent to the cloud. To fix this, AquaIntel will implement a new feature called Queue of Actions. If the user makes an action (adding a fish to the tank) while they're offline, AquaIntel will display a small notification to the user letting him / her know that the action has been queued to be synced once they are back online. This will help with data loss and keep the user informed about their digital aquarium's status at all times.

Theme 4: User Experience & Design Psychology

2.13. The "Zero-UI" Anticipatory Personalization (CleverIT UX, 2026)

- **What does it say?** Predicts that apps will move toward "Anticipatory UI," where the interface changes based on the user's past goals. Instead of making users search through multiple menu options, the interface automatically surfaces the appropriate tool at the desired time. According to the study's findings, using this approach can significantly reduce "decision fatigue" for the user and make challenging management activities much easier.
- **Implementation:** This means the app dashboard will be dynamic dashboard. The app should "surface" the most likely next task (e.g., if it's Sunday, show the Water Change guide). The app by simply moving the next task to the front.
- **Relevance:** Simplifies **Smart Management** by showing only what is relevant to the user *today*. The feeding day will show the feeding schedule first, while the maintenance day will display the filter cleaning tasks list first. This ensures users only see items applicable to the current requirements of their tank each day.
- **Improvement:** Anticipatory UI has an inherent danger in that it may be perceived as restrictive by some hobbyists who may want complete control. I envision overcoming this by including a "Dual View Toggle". By default, this app will have the "Anticipatory" UI as the primary interface when accessed by a novice. However, users will also have the choice to switch to the "Pro" UI for those with experience. This way, the user can select whether they want AI assistance by a simplified, curated pathway or have access to the complete "History Log" and raw sensor data presented in a more advanced fashion.

2.14. Micro-Tasking for Hobbyist Retention (Aquarium Adventures, 2025)

- **What does it say?** According to research from Aquarium Adventures (2025), numerous individuals leave this hobby due to feeling too overwhelmed. When someone opens an extensive "To-Do" List of any length longer than 2 hours and sees how many items are required, they quickly experience "Hobbyist Burnout." The research also shows if you split larger chores into many (more numerous) small, 5 minutes or less "Micro-Tasks", probability increases that these users will stay in the fish-keeping hobby and will have healthy fish.

- **Implementation:** Weekly chores are broken into daily "Micro-Tasks" that feel manageable.
- **Relevance:** The AquaIntel Reminder System, which is being developed by this researcher, is focused on achieving "small, yet consistent victories" in maintaining an aquarium. The app has been designed to give users only one small daily action to carry out, ensuring that they can maintain the aquarium without becoming overwhelmed by the many things that they must do. These actions are meant to turn a complex hobby into a series of quick, simple behaviours.
- **Improvement:** Micro-tasking provides ease of completion, but may become monotonous over time. To address this, I am implementing a "Rewards System." Users will earn Virtual Badges (e.g. "Master of Water Quality" or "Plant Whisperer") upon completing one month of micro-tasks, thereby providing users with a sense of accomplishment and encouraging them to return to the app to further develop their fish keeping skills through gamification.

2.15. Search Optimization in Domain-Specific Databases (Maze Report, 2026)

- **What does it say?** The Maze Report (2026) shows how users have difficulty with technical terms in applications designed for specific purposes. For example, in the aquarium hobby, many fish names can be hard to remember because they are long, use Latin, or are difficult to spell correctly. The study indicates that fuzzy searching (the ability to identify variations of similar named things, e.g., recognizing "Pleko" as "Pleco") is critical for users of aquarium apps. Many users will stop using the applications because they perceive them as faulty or challenging to use without fuzzy searching capabilities.
- **Implementation:** "Category-First" search allows users to find fish even if they don't know the specific name.
- **Relevance:** This study indicates that it is better to use a "Category-First" search rather than only using a blank search bar. The app provides users with "buckets" of information and therefore does not require a user to know an actual name before being able to search for categories that provide them with what they are looking for, for example "Tropical", "Cold Water", "Colorful", or "Small". This creates a much more intuitive way of searching through the database.

- **Improvement:** With this study, the Fish Compatibility Checker is greatly enhanced. The "entry barrier" for beginners has been greatly reduced. A user sees a fish they want, but they do not know its name, now it can be looked up by description ("yellow, black stripes") and using the logic of the Maze Report it can filter the database to find a highly possible match, allowing the user to get their appropriate care information.

2.16. Mobile UI for High-Glare Environments (Verma, 2025)

- **What does it say?** Research conducted by Verma in 2025 examines the performance of mobile applications in environments such as "high-glare". High-glare conditions are typically associated with intense brightness. It was determined that the majority of mobile applications are extremely difficult or impossible to read from near very bright light sources such as aquarium type LED lighting. In order to resolve this issue, the study recommended implementing High-Contrast Themes on mobile application screens which would provide a way for the application screens to penetrate through the glare produced as a result of the light fixture.
- **Implementation:** Another specific point made is that 'Dark Mode with Neon Highlights' (i.e. Brightly colored text, like green or blue, on a black background) is the easiest on the eye to view in a bright room. This way, the user does not need to squint and put the brightness on the phone up to 100%.
- **Relevance:** This influences the **Visual Design** of the app. As the app focuses on the aquarium owner they often use aquarium owners while standing next to the bright tank to counter the bright light and caused less discomfort to the eye it focuses on using contrast dark theme in the app.
- **Improvement:** I should not use small fonts; I must use "Large Touch Targets" because the user's hands might be wet.

Theme 5: Search & Logistical Logic

2.17. Exponential Search Logic in Unbounded Domains (Bentley & Yao, 1976)

- **What does it say?** This basic work done by Bentley and Yao introduces methods of locating information even when you don't know exactly how big the data set is. They showed that searching using an "Exponential Growth" strategy (i.e., doubling the area of

search every time) allows information to be located in the minimum possible amount of time (i.e., it determines the right "range" in no time at all, even when there are many thousands of items to be searched through).

- **Implementation:** The strategy employed can be described simply as a two step system:

Jump: The system begins by searching for the result in a progressively larger range(1,2,4,8,16) until it can determine it has found the correct range to search in-a "boundary".

Pinpoint: This enables "Expansion Logic" in your Compatibility Checker application. What this means, if you are seeking a new tank mate and in the "Beginner" groups, zero options are found within that level, the application does not terminate the search, it "doubles" the search to include "Intermediate" groups and then "Expert" groups.

Relevance: This powers the "**Expansion Logic**" in your Compatibility Checker. If a user is looking for a new tank mate and the AI finds zero results in the immediate "Beginner" category, it doesn't just give up. It "doubles" the search scope to include "Intermediate" and then "Expert" categories. This ensures the app always tries to find a solution for the user, even if it has to look into more specialized groups of fish.

Improvement: The problem with an "unbounded" search is that it could eventually produce something as stupid/deadly as 'shark' for 'goldfish bowl', because that's the 'only' result. In order to overcome this in AquaIntel, I will implement a 'Threshold Cap'. This is a safety limit. No matter how 'broad' a search becomes, it can never go above a certain number of 'Red Lines' like the huge disparity in temperature, or the predator/prey situation.

2.18. Community-Sourced Compatibility Data (AquariumStoreDepot, 2026)

- **What does it say?** In a study that surveyed tens of thousands of "real world" Community tanks, AquariumStoreDepot (2026) found that existing compatibility charts do not prove true. Many "risky" pairings of fish will actually get along because the plants completely interrupt the "line of sight" allowing timid fish to have somewhere to retreat to and stopping the more dominant fish from owning the whole tank.
- **Implementation:** This experiment verifies "Visual Cover" (plant life, wood and rocks) is the unknown variable influencing successful cohabitation. For the AquaIntel logic, this

means fish species alone cannot be the predictor of whether two fish species are compatible and that the application would need to take the aquarium itself (physical layout) into account.

- **Relevance:** This is another dimension the Fish Compatibility Checker will add on. The AI Assistant will now question about how planted the tank is before giving a score: "Is your tank heavily planted?". If the answer is yes, the AI will be able to "unlock" some of the pairing categories which are labeled "Caution" in a less planted tank.
- **Improvement:** Just saying, "add plants" doesn't really do a beginner much good when they have no clue as to what plants are going to be best at providing cover. I would like to make this better by having the "Planting Guide" included in the AI Assistant. So instead of a 'yes' or 'no', the app would be designed to read "These fish will be compatible IF these three tall plants are added to create an adequate visual barrier." This allows the AI to be a designer that assists the user.

2.19. Behavioral Ethics in AI Advice (CleverIT Research, 2026)

- **What does it say?** CleverIT (2026) carried out research into the responses of people when receiving advice from AI and discovered that they rebelled against "Forceful Advice"-sentences telling users what they 'must' or 'have to' do. They were far more likely to act on instructions and believe the app if it provides "Evidence-Based Suggestions" as this change in tone is less imposing and appears to have shifted from an impersonal figure of authority to more of a wise peer.
- **Implementation:** Another possible wording modification to enhance the conversational tone. Rather than the AI state " You need to change the water today" it is proposed to use less authoritarian wording supported by the research as in, "Research indicates a 20% water change today would best keep your Nitrates below a harmful level". This gives the user the confidence they are making a correct decision rather than just an order.
- **Relevance:** In this section, we discuss the "Personality" of the AquaIntel Assistant. It helps the app be a reassuring friend for the user and a tool for their hobby. It is especially critical to take this approach with novice aquarium keepers, many of whom are quite frightened by the chemistry and science involved in fishkeeping. A friendly, scientific approach can

go a long way in keeping the user at ease and confident even when their tank is showing issues.

- **Improvement:** To take this to the next level I will include a "Show Source" button alongside any "critical" advice the AI gives. For instance if the app advises giving a specific type of medicine, or advises on a particular water level adjustment the user will be able to press a button to see where that advice comes from. (E.g., "According to Sandford 2024 and current Ammonia levels"). This will combat the "Black Box" problem of AI and demonstrate to the user that the advice is not arbitrary.

2.20. Platform Consolidation in mHealth & Utility (PLOS, 2025)

- **What does it say?** A study in the journal PLOS Digital Health (2025) investigated the reasons people abandon niche apps. They found that 43% of users uninstall because of "Cognitive Exhaustion". This occurs when a single user must juggle multiple apps (e.g. A calendar app for chores, an internet browser to search for fish species names, a notes app for water quality test results) in order to accomplish a single task. "Managing multiple disjointed tools is enough to make the user give up altogether".
- **Implementation:** The study suggests the use of a Unified Dashboard strategy. Rather than forcing the user to switch through different "modules," the app should bring Checker, Assistant and Reminders into one single screen. This "All-in-One" view would reduce the "mental switching cost" and make the app appear more like one brain than several folders.
- **Relevance:** This study has successfully confirmed the basic design of AquaIntel. Because you have kept the Compatibility Checker, AI Assistant, and the Maintenance Reminders under a single module, you are directly resolving the issue of "Cognitive Exhaustion." There's no need for the user to know where they should go to get a response, they just open the app and the answer is in front of them.
- **Improvement:** However, to best address fatigue, I will utilize a "Two-Tap Rule." The interface I design will also ensure it's not possible to "Menu Deep-Dive" – if something is a "core" feature (it's not possible to access a core function from the home screen in more than two taps. This keeps the app quick, simple, and 'light' for the user to use, especially if the user is frantic about their tank.

3. Primary Research

Primary research is a key part of the AquaIntel project. It offers direct insights into the real-world challenges aquarium users face. While secondary research provides theoretical knowledge and existing solutions, primary research focuses on collecting first-hand data from actual users. This helps us better understand their behaviors, needs, and expectations.

The goal of primary research in this project is to identify common difficulties in aquarium management. These include maintaining water quality, managing feeding schedules, and understanding fish compatibility. It also aims to assess how aware users are about proper aquarium care and the use of digital tools for help.

By gathering this data, the project ensures that AquaIntel is not just a theoretical system but a user-centered solution that tackles real problems. The findings from this research directly shape the design, functionality, and features of the system, ensuring it is relevant and usable for the target audience.

3.1 Objective of primary research

The objectives of the primary research of the project AquaIntel

- To find the problem faced by aquariums owners
- To know the user knowledge and awareness
- To examine the current aquarium management
- To assess the utilization of current tools and Technologies

3.1. Market research (existing products)

Aquarimate: The Main Value Statement: The Total ArchiveAquarimate is created as a professional logging tool. Its advantage is that it has historical data and is multi-device-synced. **Aquaribase:** It contains one of the largest livestock collections in the industry (thousands of species) of both freshwater and saltwater. **Visual Timelines:** One of the most notable features is the ability to use photos on a particular fish or coral and ensure growth or recovery after an illness across months or years. **Hardware Integration:** Hardware integration has been recently added to enable it to import water test results such as Apex Trident NP and Hanna Marine Master. **The Obstacle of the Amateur:** "The cockpit of the Pilot" The greatest objection to Aquarimate is that it is like filling out your taxes. **Cognitive**

Load: The dashboard is filled with spreadsheets, graphs and sub-menus with deep sub-menus. As a novice struggling to ensure that your first Betta is not killed, it is overwhelming. No Why behind the Data: It informs you that your pH is 8.2, but lacks the tone of supportive AI and evidence-based suggestions that AquaIntel aspires to deliver. It is a calculator and not a coach.³ Pricing & Business Model The Paid Wall: Unlike most contemporary apps, which have a strong free version, Aquarimate is a commercial product. It typically involves a one time purchase (approximately 9.99) and a subscription to Aquarminute Live (cloud syncing) of 9.99/year. Friction: When their hobby is new, beginners are usually hesitant to spend money on buying items until they get to its first week when they are unsure of whether they would continue doing it.

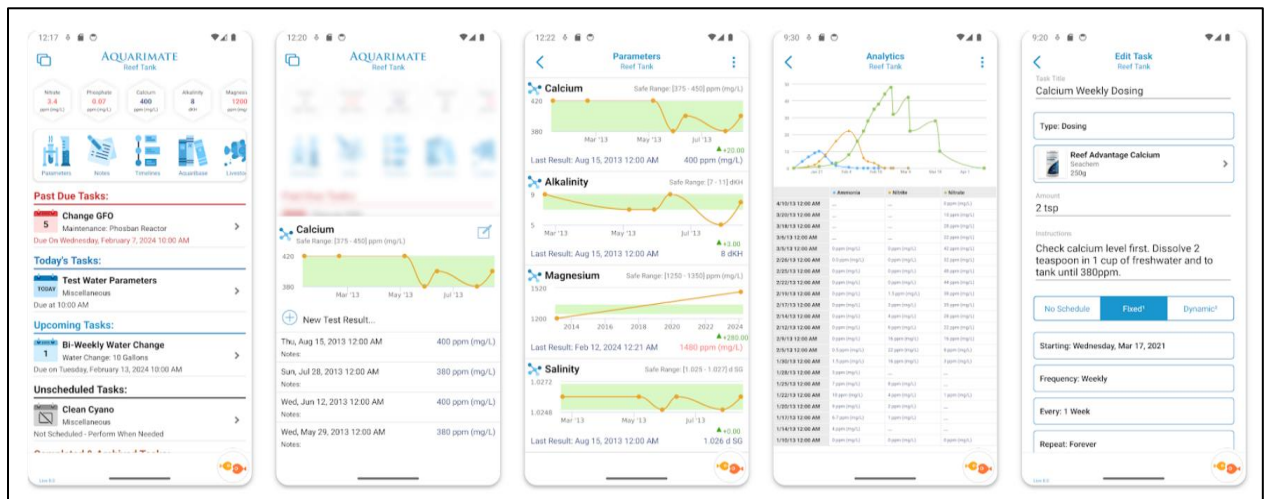


Figure 2 Aquarimate UI

Aquarium Note: . The Precision Logger Core Value Proposition.

The advantage of this application is that it primarily deals with technical tracking. It is constructed to suit individuals who wish to observe the figures and make certain of their water chemistry being ideal. Complete parameter monitoring: It monitors 24 water parameters (pH, Nitrates, Salinity, Calcium, etc.).

Growth & Age Tracking: It also has an in-built counter unlike other apps that informs you of the number of days spent by a fish or plant in your tank.

Inventory & Expense Management: It enables users to record the "Goods" (filters, food, salt) and the overall cost of the hobby is automatically computed, which is both convenient and a little frightening to the hobby lovers!

2. The User Experiences: "Usefulness and not Beauty"

The 'Legacy' Feel: The interface of the initial Aquarium Note is somewhat reminiscent of an older Android application. It is extremely quick and consumes practically no battery, yet it does not have the appearance of more recent applications

Manual Entry: No AI/automation is involved. All this has to be done manually. This is fantastic in terms of precision, but causes the so-called Cognitive Exhaustion of which we spoke in Theme 20.

Trustworthy Backups: It has a notorious easy export of HTML/XML, which means that the user will be able to keep their data despite the abandonment of the app.

3. The "AN1" vs "AN2" Split

In 2011, the developer made Aquarium Note 2 (AN2) available to update the experience.

Modern Dashboard: AN2 has a far more clean, translucent interface, and appears more of a 2026 application.

Task Management: AN2 has become closer to the Micro-Tasking concept with additional powerful To-Do lists and reminders (Theme 14).

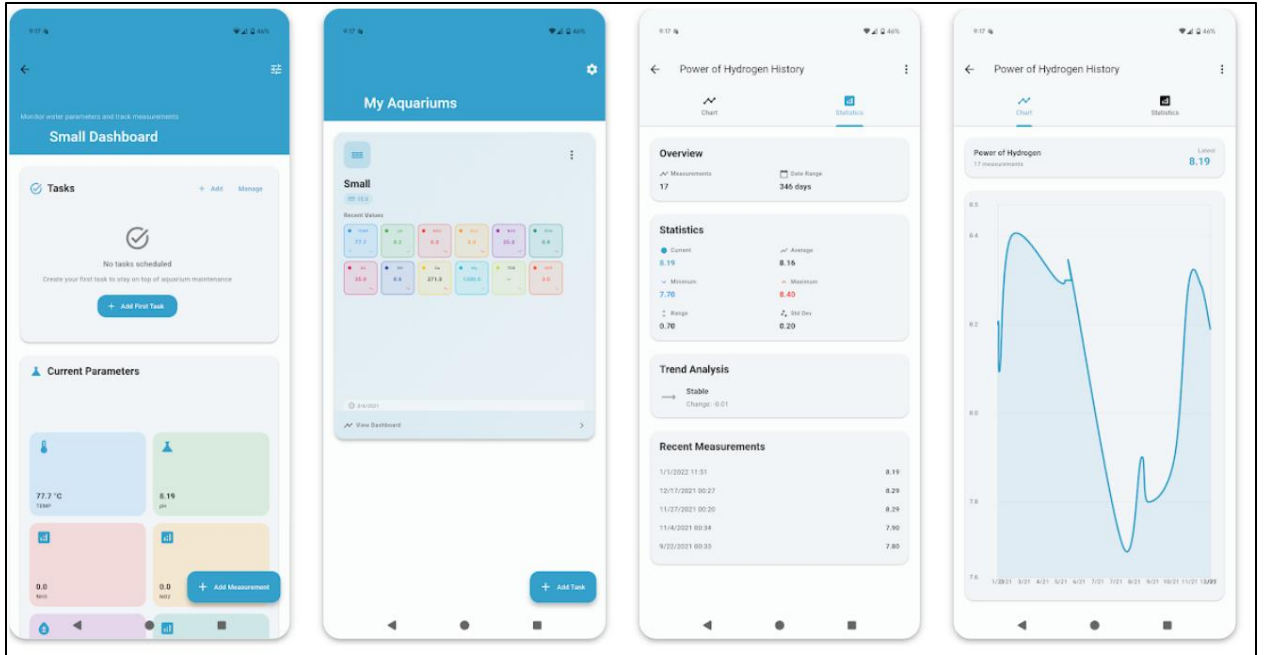


Figure 3 Aquarium Note UI

All these app established in the market are huge in user base though they lack some features in terms they fully some are not good in giving good user interaction. Likewise, some are expensive more beginners and complex and the same time. AquaIntel is an application designed to fulfill the gap created by the other applications by combining the missing things of others application and better pricing and less complex.

3.3 Data collection and analysis

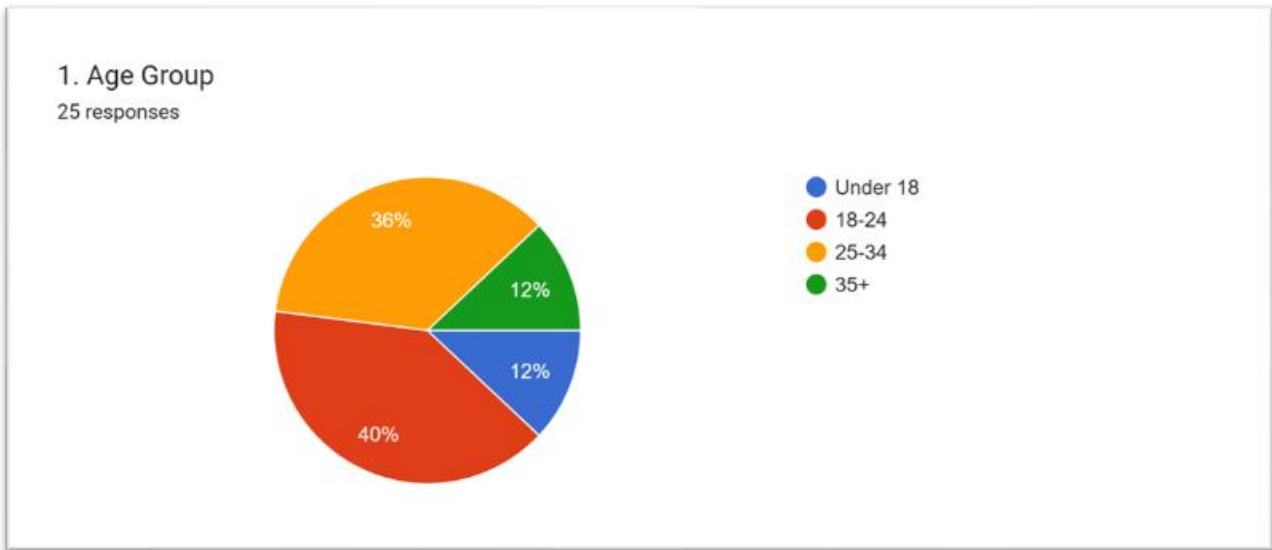


Figure 4 age group

The majority of our audience is split into the 18-24 (40%) and 25-34 (36%) groups. This confirms your primary users are Gen Z and Millennials who should be tech-literate and happy to use mobile apps for managing their hobbies. A digital first product such as AquaIntel therefore suits the profile perfectly.

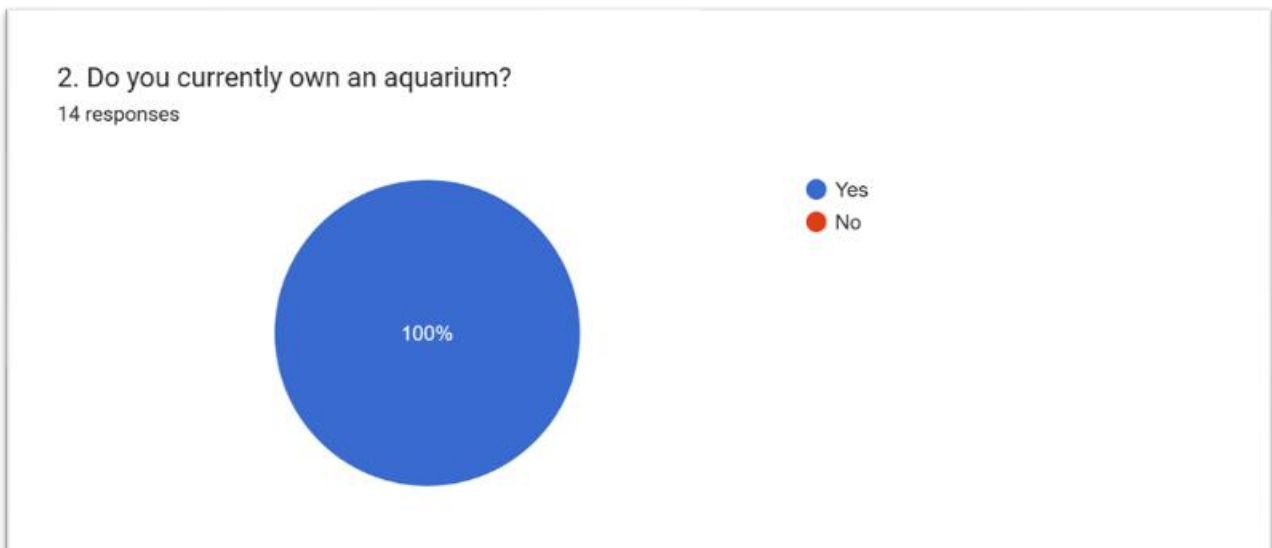


Figure 5 Aquarium

This result shows a 100% ownership rate among the 14 respondents for this specific question. This is excellent for your initial data collection, as it means every person answering has firsthand, current experience with the daily tasks and frustrations of maintaining a tank, making their subsequent feedback highly reliable.

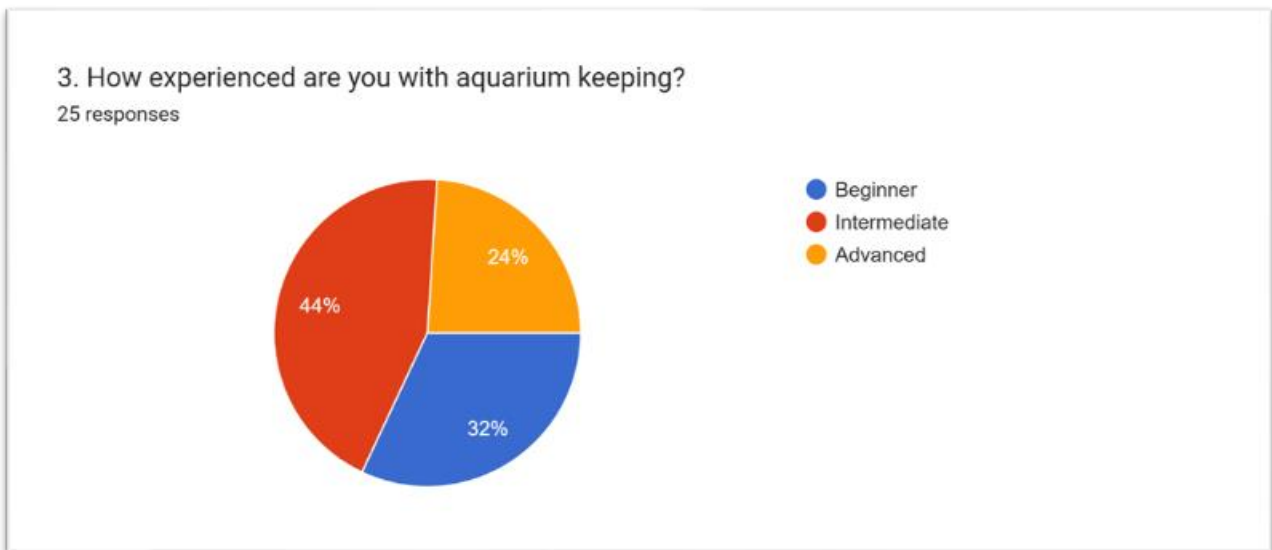


Figure 6 Experience in aquarium keeping

A good balance of skill is apparent from the user segmentation. There are 44% Intermediates (who are the largest group) and then 32% Beginners. As it is almost a third of users are beginners, having the app accessible, and being able to "fuzzy search" the identification will be incredibly important for those newer to the hobby.

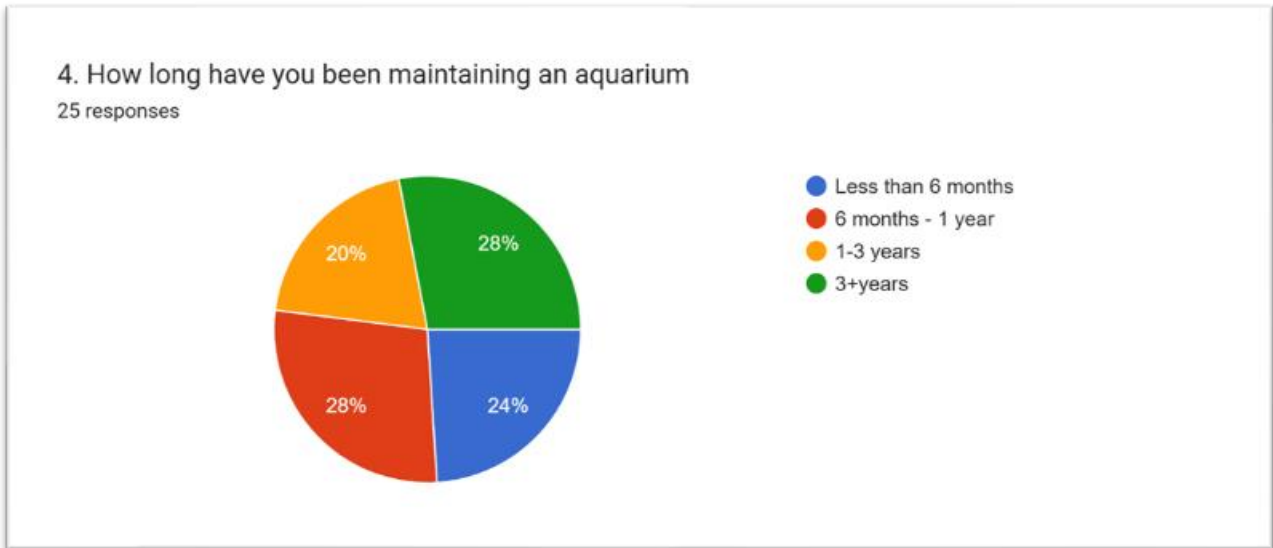


Figure 7 Maintaining an aquarium

The level of experience is fairly split; 52% of the user base have one year or less of experience while 48% have been practicing the hobby for more than a year. This chunk of 'newcomers' is the high-risk segment for "hobbyist burnout", so you can definitely see why there is a need for the maintenance reminder system that you are proposing.

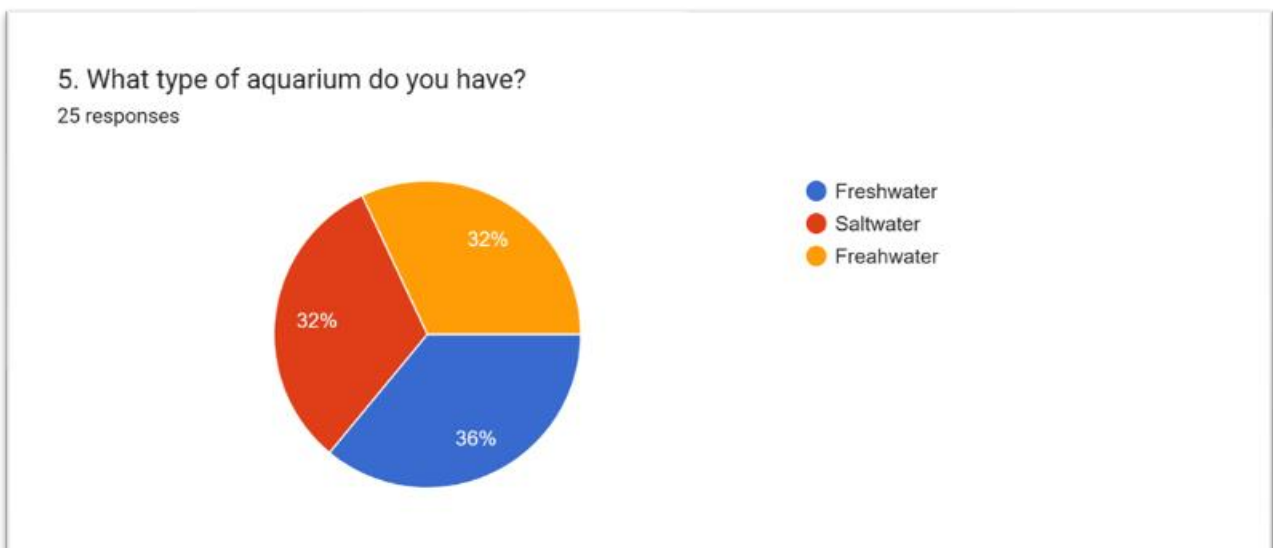


Figure 8 Aquarium Type

The data shows a roughly equal split between Freshwater (approx. 68% combined) and Saltwater (32%). Note: There is a typo in the survey ("Freahtwater"), but combined, it shows freshwater is the dominant interest. AquaIntel will need to support both, though freshwater features may see slightly higher engagement.

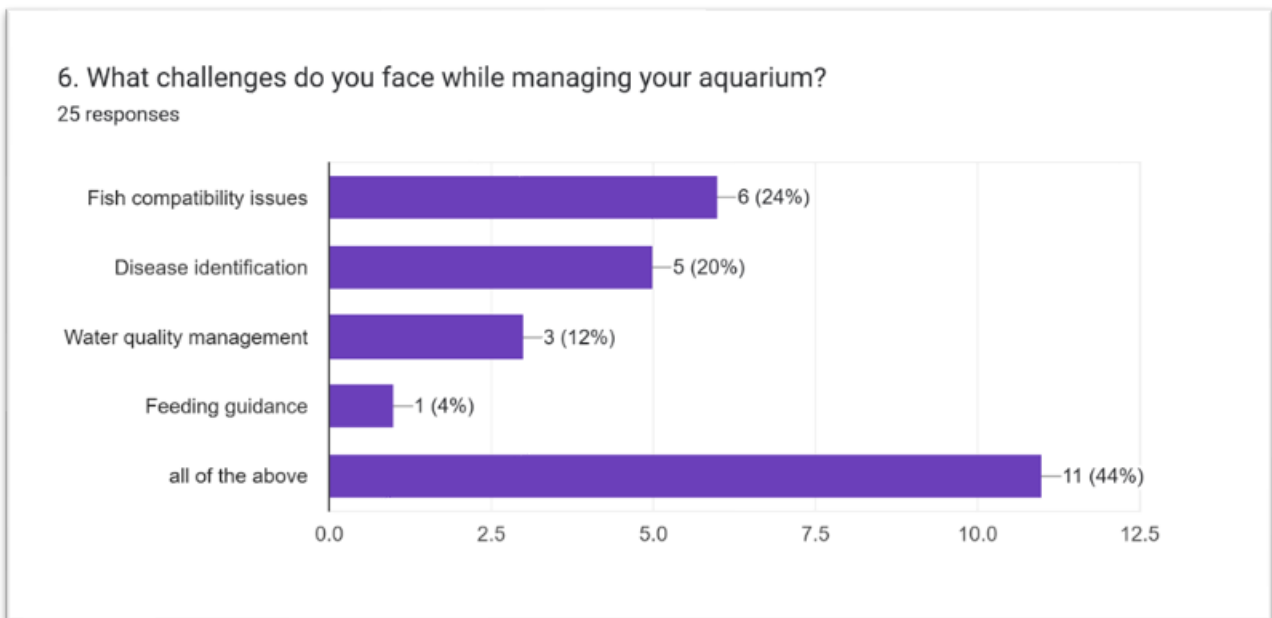


Figure 9 Challenges faced

The level of experience is fairly split; 52% of the user base have one year or less of experience while 48% have been practicing the hobby for more than a year. This chunk of 'newcomers' is the high-risk segment for "hobbyist burnout", so you can definitely see why there is a need for the maintenance reminder system that you are proposing.

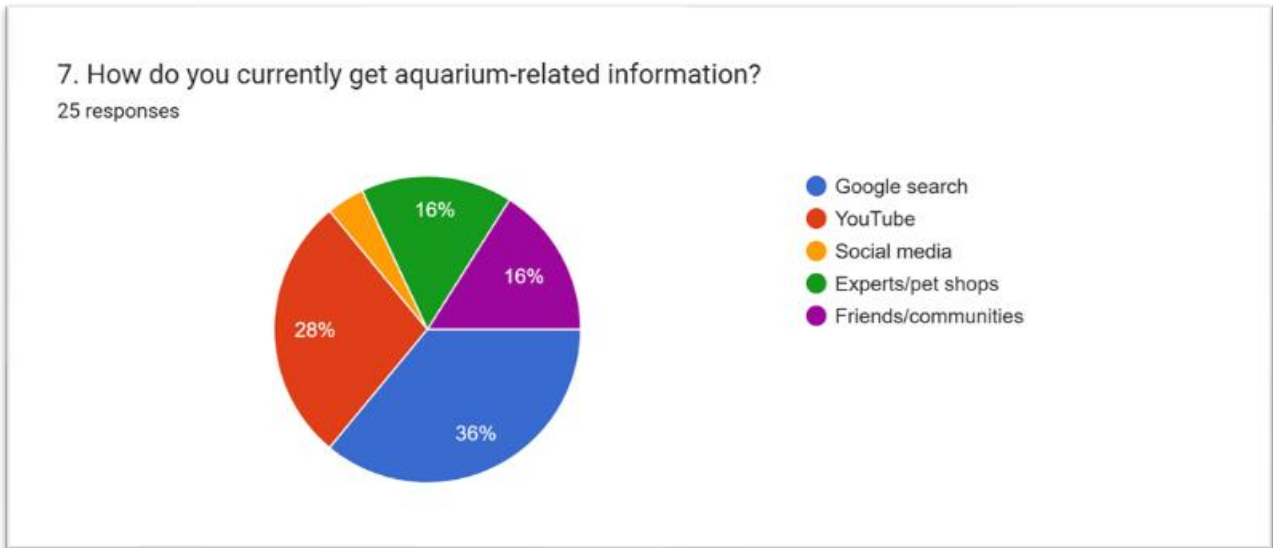


Figure 10 Information related aquarium

The main platforms people access this information are google search (36%) and YouTube (28%). These numbers indicate that people are willing to search, but the knowledge isn't put in one convenient place. AquaIntel can fill this role.

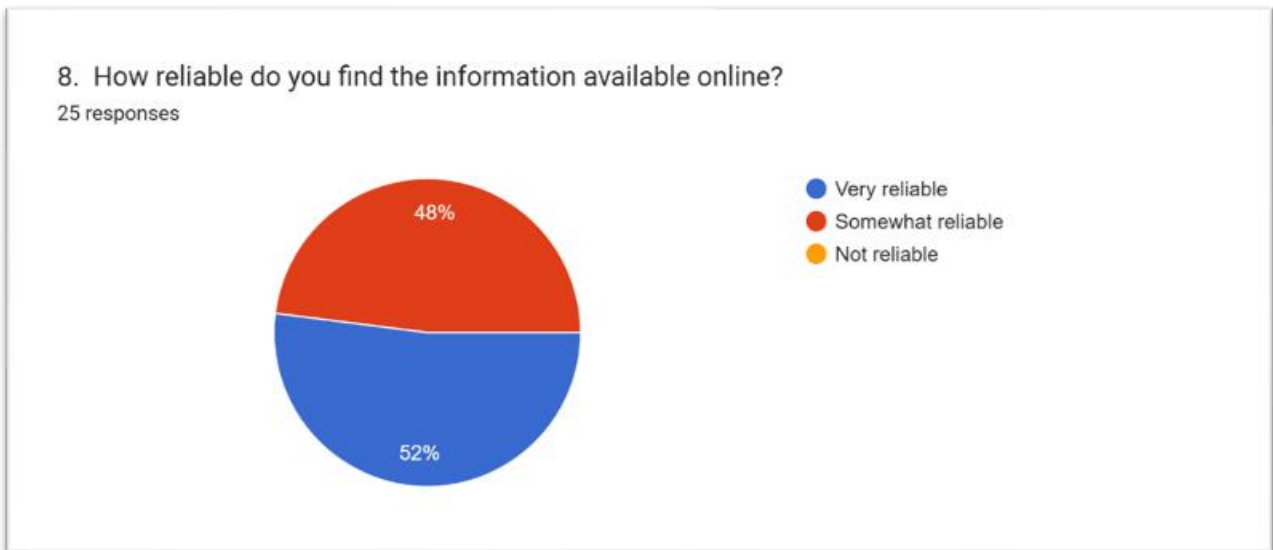


Figure 11 Online information reliability

The gap is fairly even between those who perceive the web as Very Reliable (52%) and Somewhat Reliable (48%). The "Somewhat Reliable" grouping suggests that a level of distrust is being placed

on existing search results, which creates space for AquaIntel to brand itself as a "research-backed," and therefore, trustworthy, source of information.

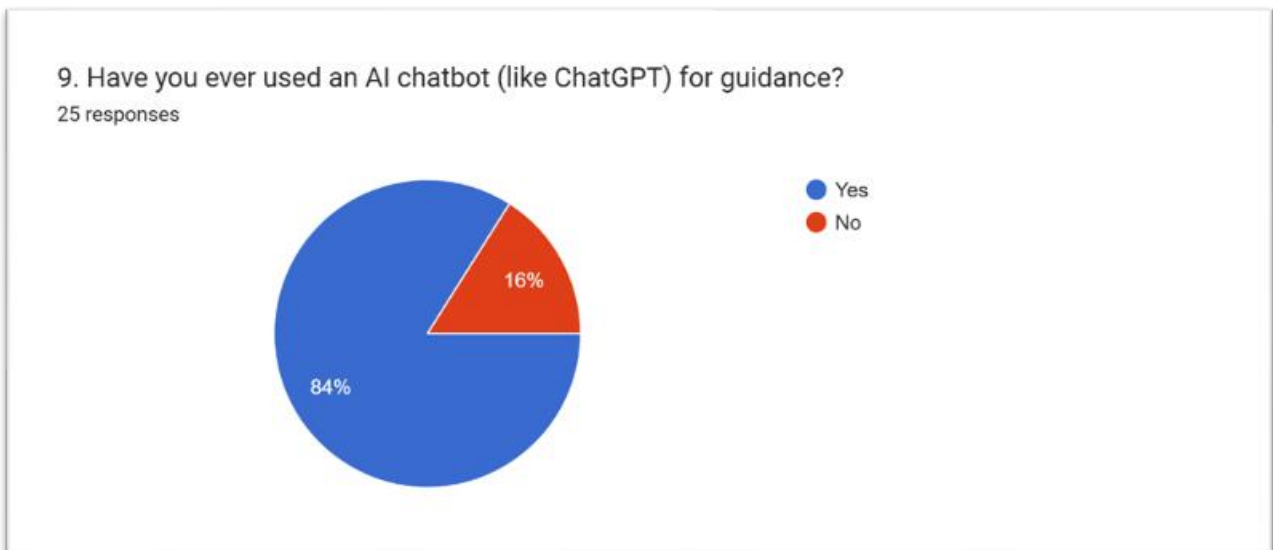


Figure 12 Used ChatBot

A good 84% of our target audience have previously interacted with an AI chatbot like ChatGPT for advice. This is an excellent sign for AquaIntel, your target audience is familiar with the use of AI and more receptive to the AI assistant within your application.

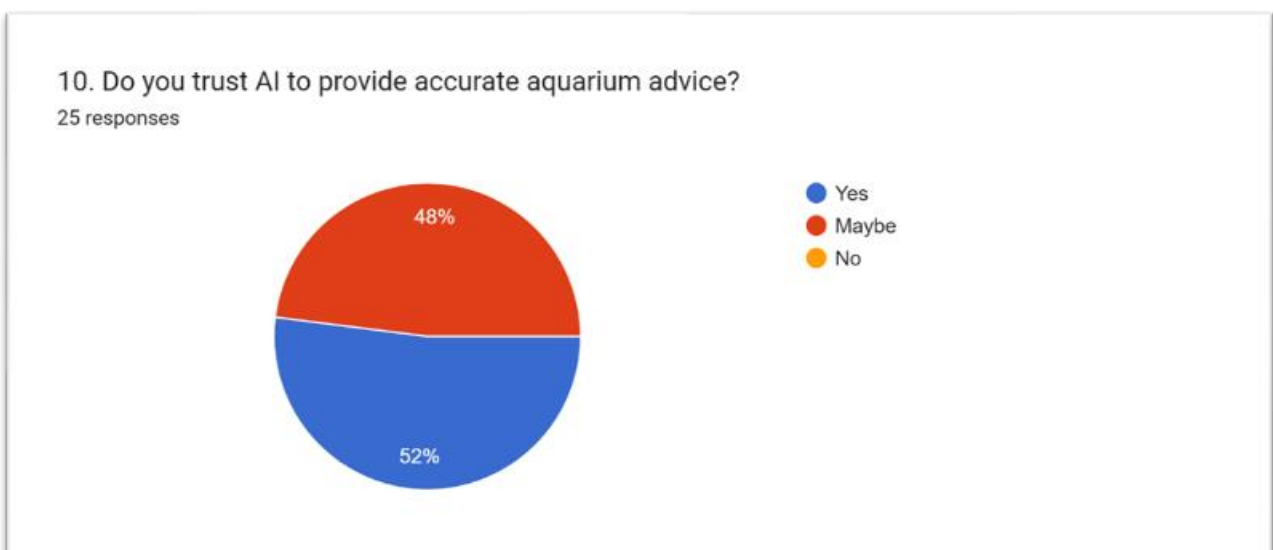


Figure 13 Accurate aquarium advice

Although 84% utilize AI, 52% have full confidence in it, with 48% responding with Maybe. It is an important lesson: users are willing to use AI but doubt its accuracy. In order to succeed, AquaIntel needs to highlight that its AI is trained in particular, validated aquatic databases to persuade the doubtful group of people in the Maybe category.

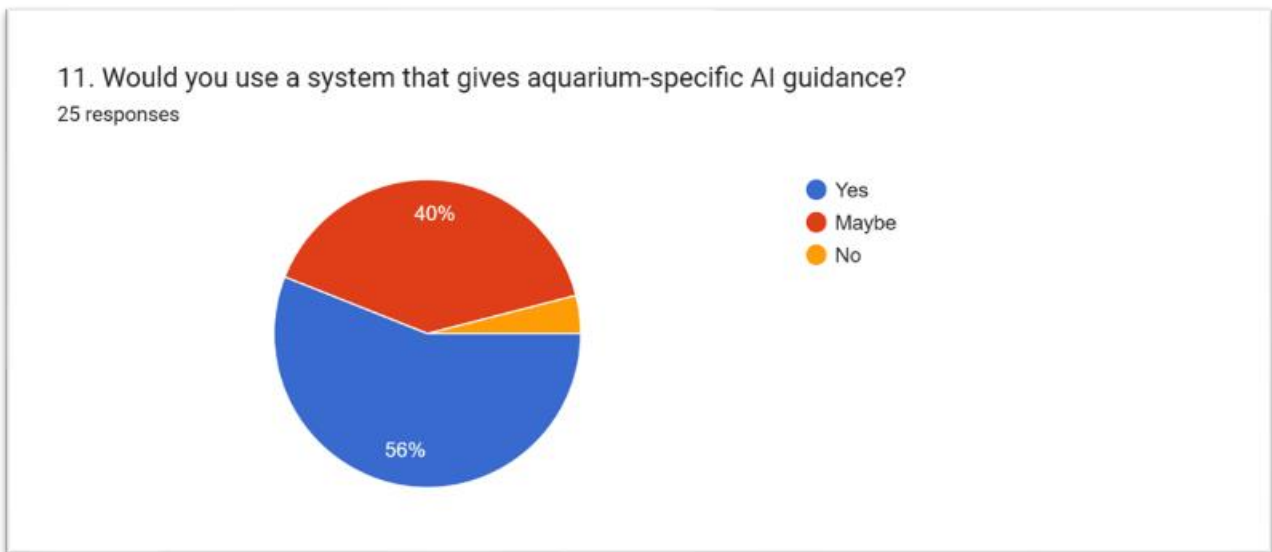


Figure 14 Aquarium-specific AI guidance

Although nearly half of users desire the entire package (40% All of the Above), the most popular single feature is the AI Chatbot (36%). Next are the Fish Compatibility Checker (24%), and the Aquarium Dashboard (24%). This strengthens the argument that the conversational AI and the compatibility database are your hero features, although a multi-tool app would be preferred.

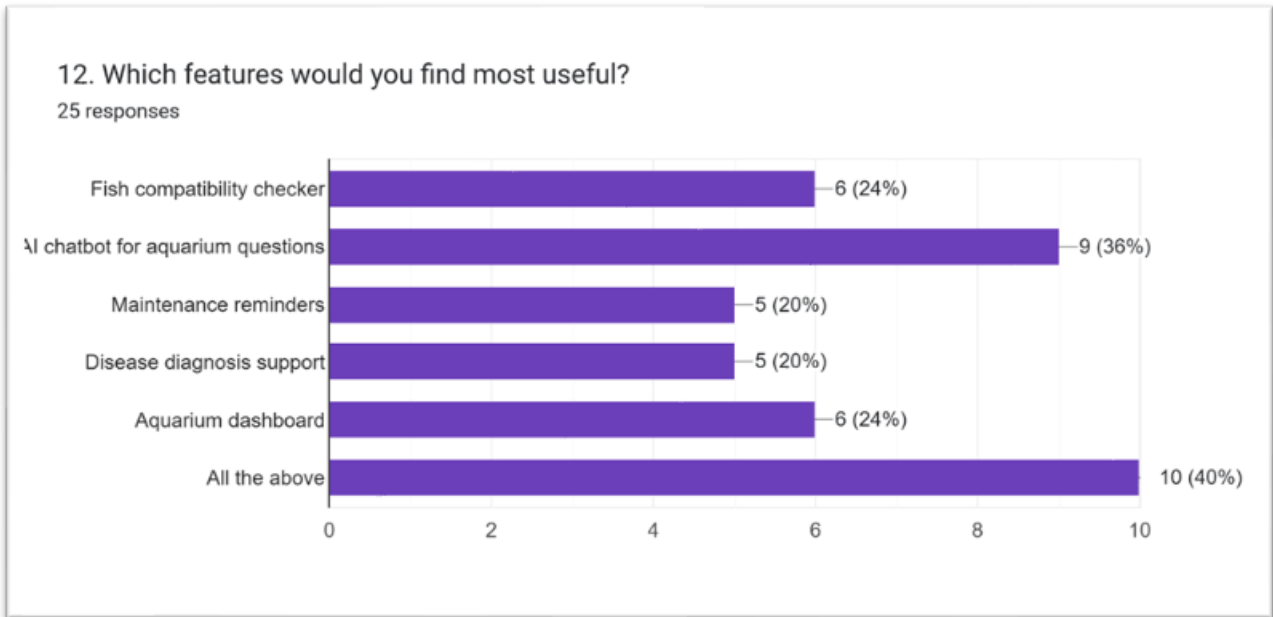


Figure 15 Most useful feature

The most one-sided finding of the survey is that 96% of the participants would choose a system that only deals with aquarium knowledge rather than a general AI. This is a titanic recommendation of the Intel section of AquaIntel- users do not desire a generic bot, they desire an expert system that speaks fish.

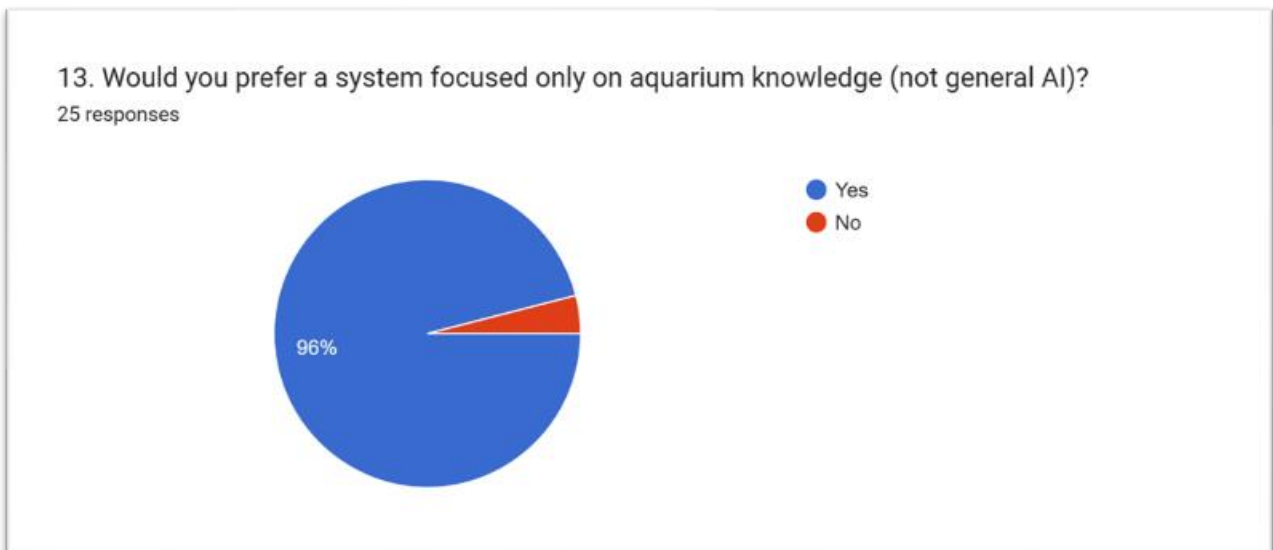


Figure 16 Aquarium focused system

The users demand accuracy: 100 percent of those surveyed rated correct, advice specific to domain as either Important (60 percent) or Very Important (40 percent). This proves that the success of AquaIntel will depend a lot on the quality and review of the information that drives your AI.

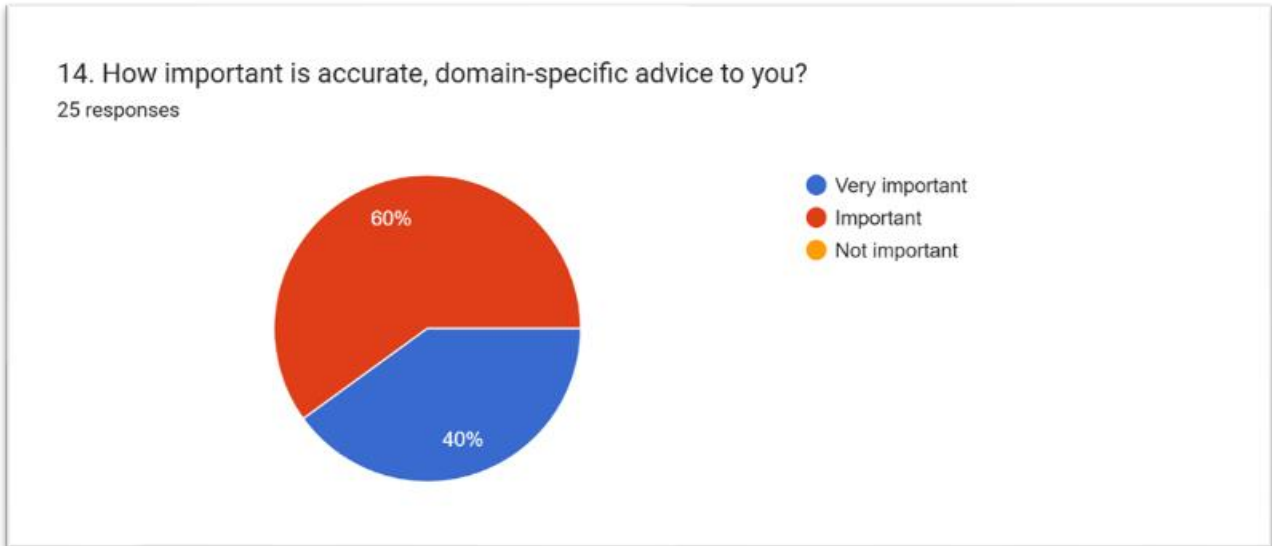


Figure 17 Domain-specific accurate

Most users (68) like a Mobile App and the rest are divided into Web and Both. This is a strong indication that mobile-first should be your first development that will enable you to have AquaIntel in the hands of your users as they are literally standing in front of their tanks.

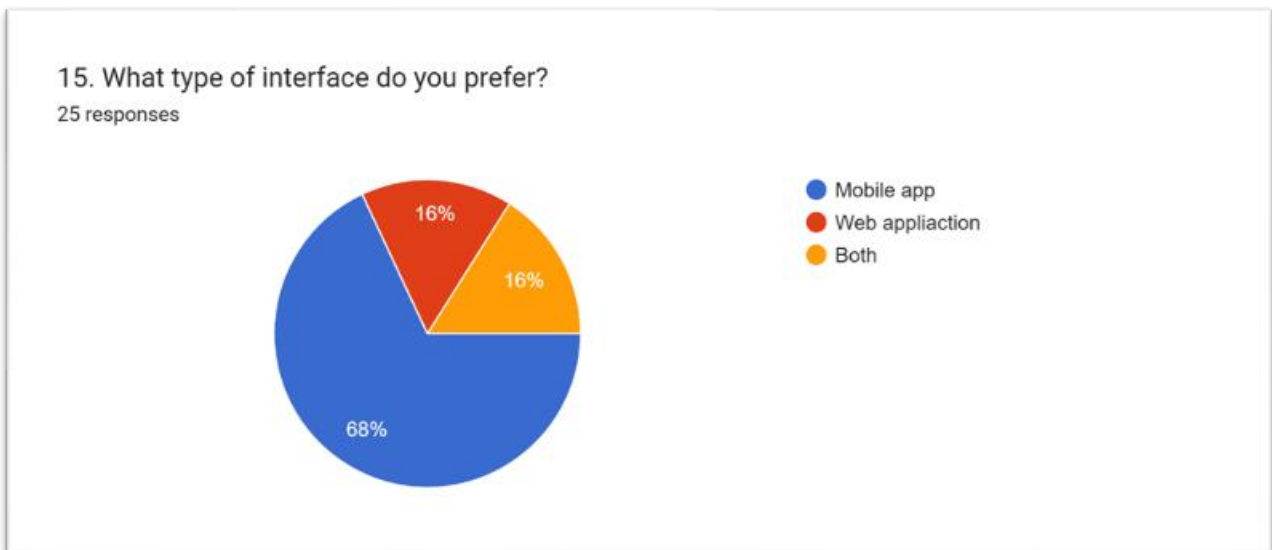


Figure 18 Interface preference

User experience (UX) is a priority, and 50/50 between very important and important. None of the respondents thought that ease of use was not important. That's why you should pay attention to accessibility-first design and fuzzy search to make the app not another complicated task.

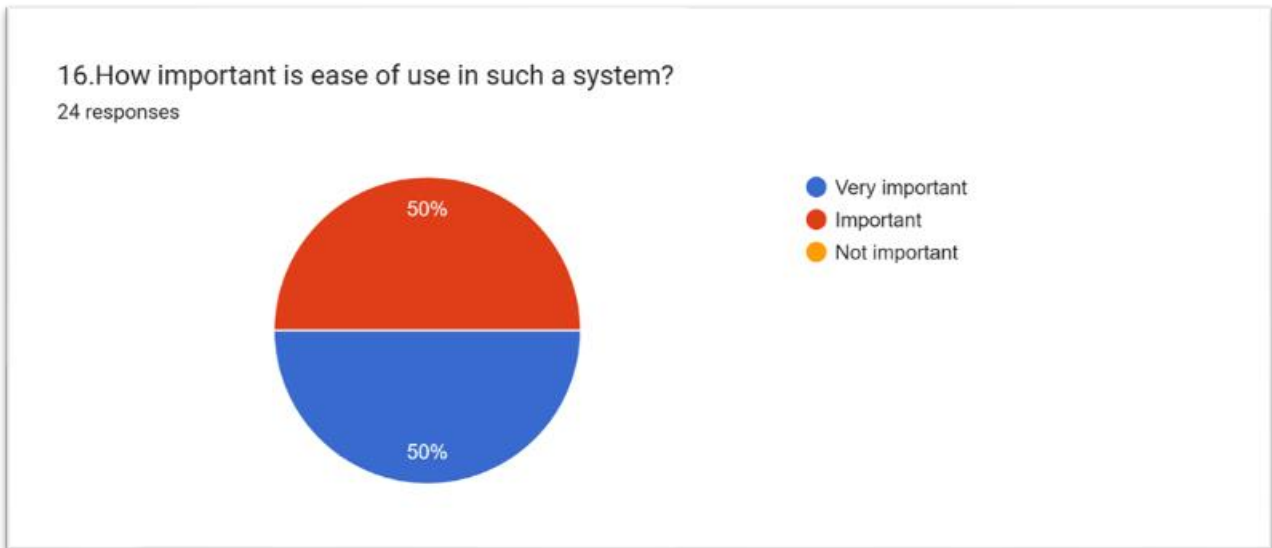


Figure 19 ease of such system

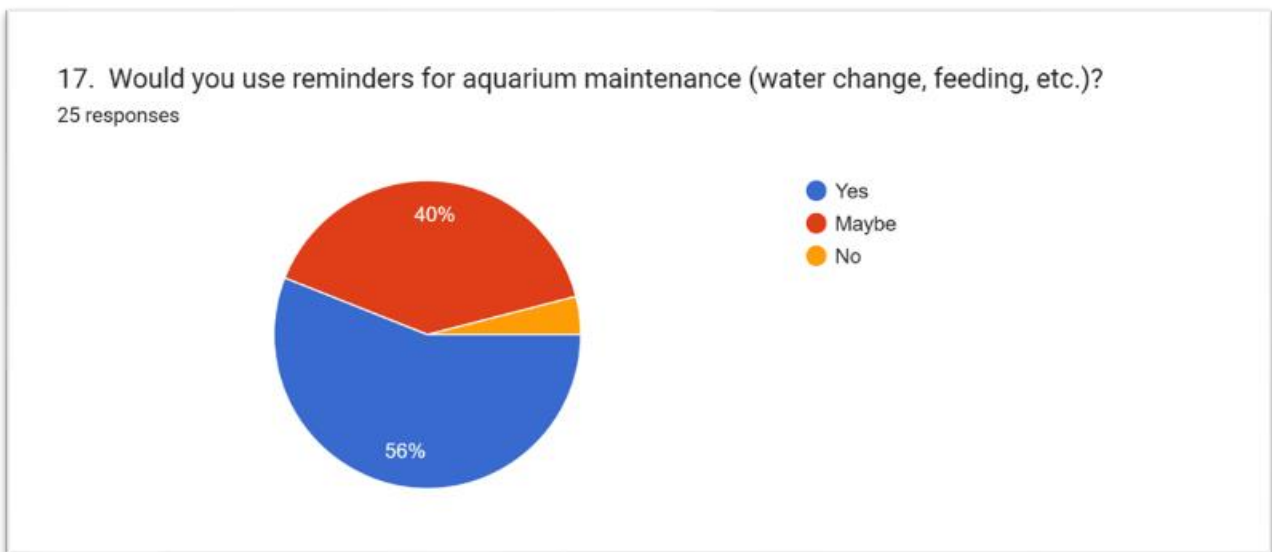


Figure 20 Reminder

The desire to have automated assistance is high: 56% have answered Yes, and 40% have answered Maybe they would like to have reminders about changing the water and feeding. This underlines

your theory that a well-organized reminder system is one of the main solutions to hobbyist burnout prevention.

3.4. User Insights

Quantitative Analysis

Measuring the Market. The planning phase is supported by quantitative research. Through the gathering of numerical data in our community survey we will be able to discern evident patterns and trends in the hobbyist arena. In the case of AquaIntel, this information will enable us to eliminate guesswork and measure the demand on certain features. An example is the fact that 96 percent of users would rather have a domain-specific AI with a general one, which gives us a high-confidence requirement on our technical architecture. It is this objective data, that is critical to prioritizing our development roadmap, as it shows precisely where the biggest of the pain points are in volume and frequency.

Qualitative Analysis:

Numbers can tell us how big a problem is, but qualitative research enables us to know the human experience of those numbers. This stage takes us a step further to explore the why of user frustrations, say, why 48 percent of hobbyists only somewhat trust what they find online or what exactly triggers anxiety resulting in hobbyist burnout. This study will give the study of open-ended responses and behavior patterns so that our interface is not only functional, but sensitive to the user experience. In the case of AquaIntel, qualitative insights can inform our UX/UI design, making our 5-minute maintenance wins and Fuzzy Search capabilities to be truly intuitive and helpful to the psychological well-being of a beginner.

This chapter is about planning and developing the project. It talks about how we did the project from start to finish. We used a way of doing things to help us make the system. We gathered information from people and from things that already existed. We used two kinds of methods to get this information.

The project had a plan for how it would look and work. We also had a plan, for how we would put everything and test it. We wanted to make sure the final project was good and worked well. We used tests to check the project and make sure it was what we wanted. The project had to be effective and reliable.

4. Project Plan

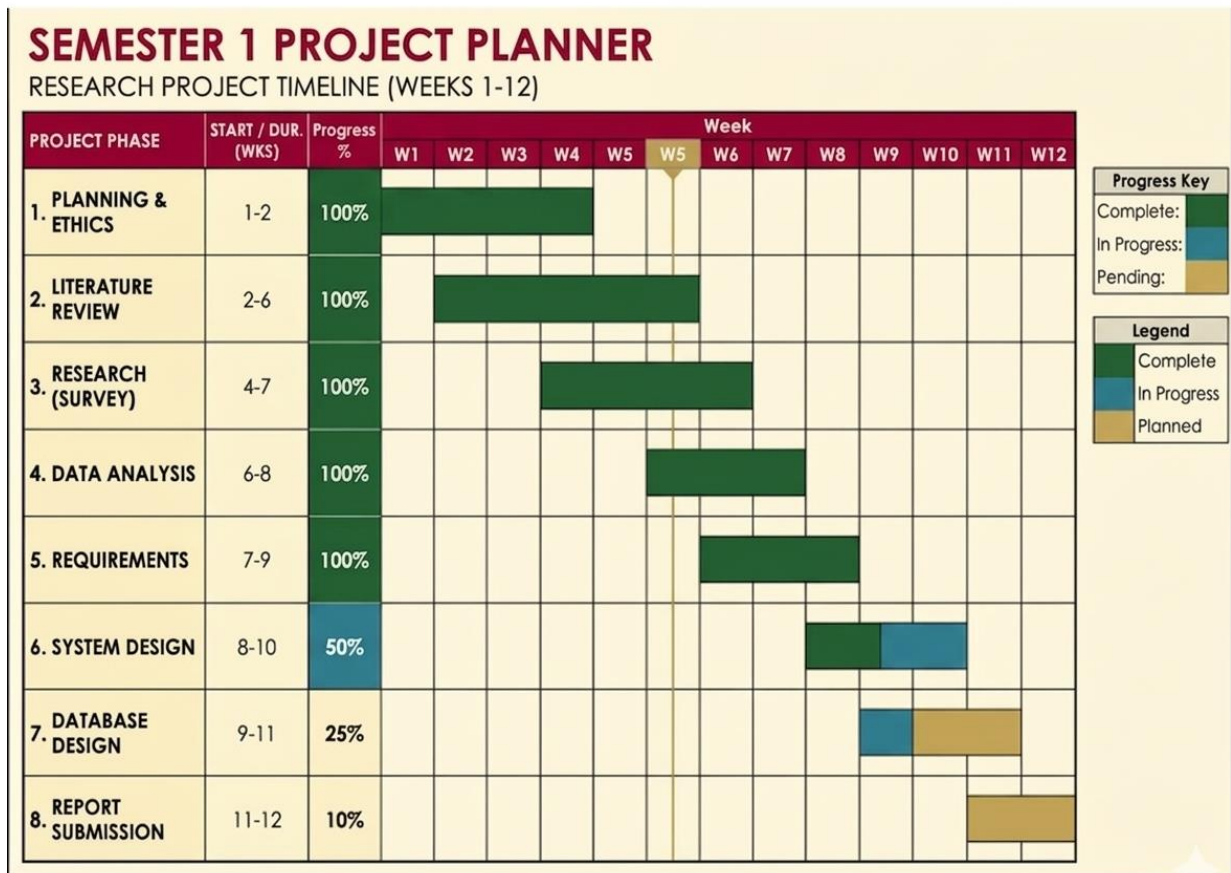


Figure 21 Ghant Chart for Semester 1

SEMESTER 2 PROJECT PLANNER

RESEARCH PROJECT TIMELINE (WEEKS 1-12)

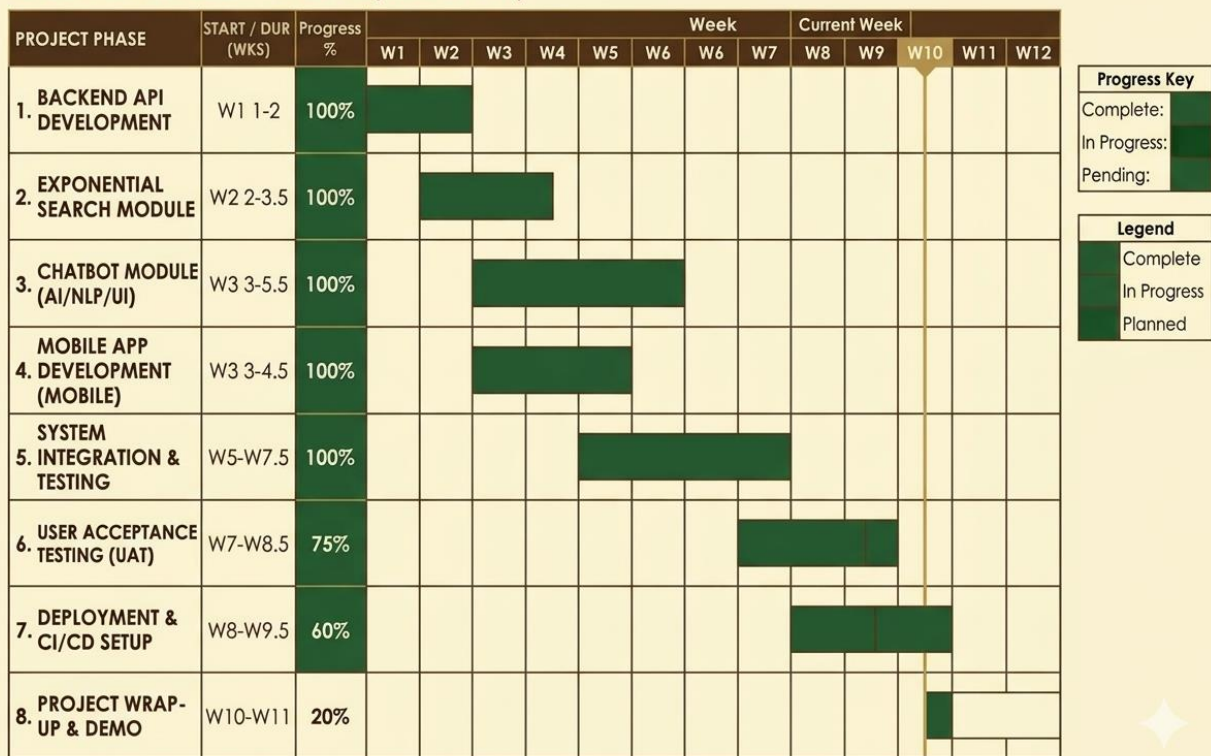


Figure 22 Ghant Chart for Semester 2

4.1. Risk Analysis

Risk Analysis: a Pre-Flight Safety Check of your app. We must look down the road before we begin to write any code and see the potholes in the road so we can dodge them. AquaIntel involves working with living animals, and therefore, any error by the AI can be critical. This reflection is our method of being a responsible inventor, what is likely to fail, the probability of this happening and how we can construct safety nets to ensure that the fish remain healthy and the app is reliable. We are not wishing well but making plans to succeed by planning on the worst.

Risk ID	Risk Description	Likelihood	Impact	Severity	Mitigation Strategy	Color Code
R1	AI chatbot gives irrelevant or incorrect responses	Medium	High	High	Improve prompt design, restrict domain strictly, test with real queries	
R2	Difficulty in restricting AI to only aquarium-related topics	High	High	Very High	Implement keyword filtering and domain constraints	
R3	Notification/reminder system fails or delays alerts	Medium	Medium	Medium	Use reliable scheduling libraries and test thoroughly	
R4	Data loss or improper database handling	Low	High	Medium	Use proper database design, backups, and validation	
R5	Integration issues between frontend, backend, and AI module	Medium	High	High	Follow modular development and continuous integration testing	
R6	Poor user experience due to complex UI	Medium	Medium	Medium	Conduct user testing and simplify UI design	
R7	Limited training data for aquarium-specific AI responses	High	Medium	High	Use curated datasets and predefined knowledge base	
R8	Time constraints affecting project completion	Medium	High	High	Follow strict project timeline and prioritize core features	
R9	Technical bugs during development	High	Medium	High	Perform regular testing and debugging	
R10	Lack of user engagement or adoption	Low	Medium	Low	Improve usability and provide helpful features	

Figure 23 Risk Analysis

4.2. Risk Heat Map

Heat Map

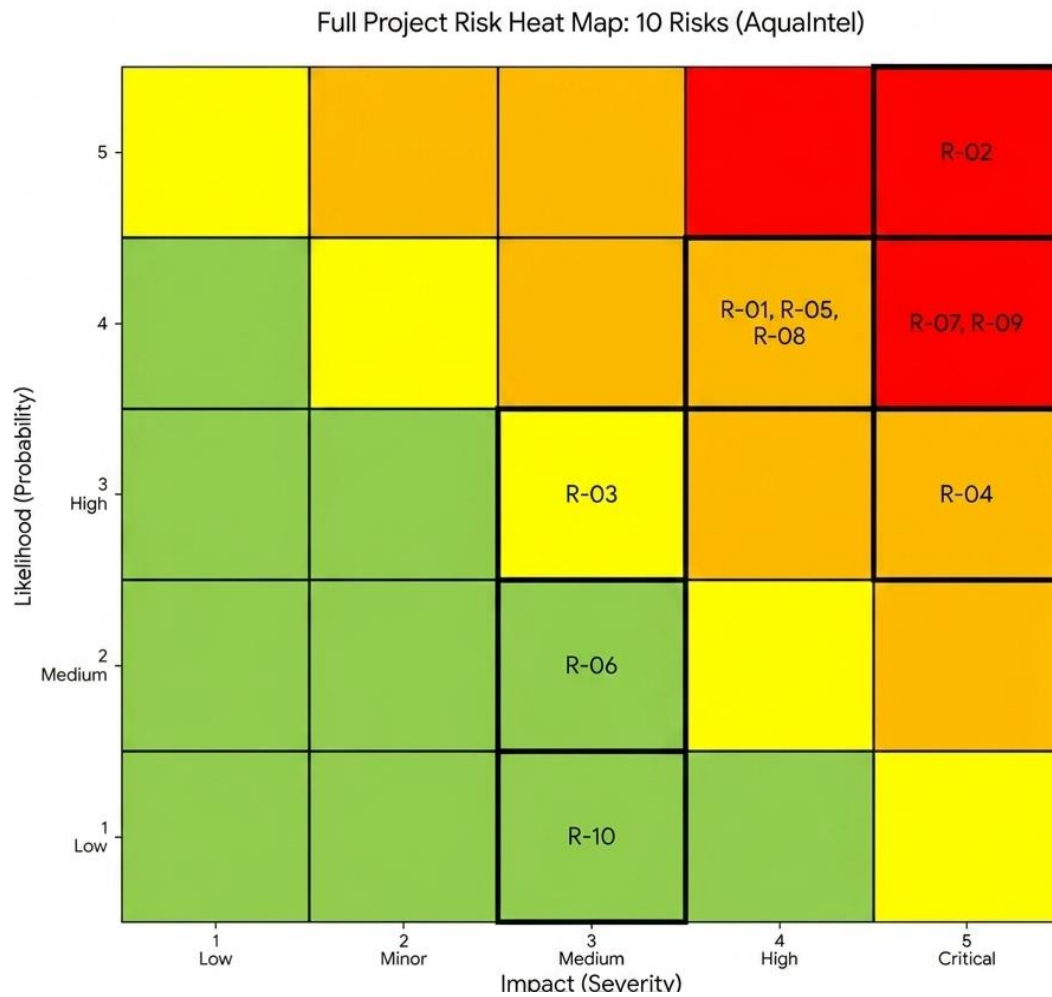


Figure 24 Heat Map

5. Artefact Planning

5.1 Methodology

Methodology is the guideline that a development process follows to create an ideal system. The methodology we choose is really important when we are making projects. We need to have a plan so that we can get everything done on time and make sure it works well. The right methodology helps us finish the project quickly avoid problems and make sure users are satisfied. For AquaIntel we are using the Agile Methodology. Agile is an iterative and flexible approach that allows the system to be developed in small modules with continuous testing and improvements. This also allows the system to deliver working features in stages. This makes it easier to track the progress of the system and ensure each component of the system functions before moving forward with the system. Encouraging feedback from people, it helps boost collaboration, making it easier to find and fix problems early on. This is highly suitable for AquaIntel as it includes features like AI chatbot, reminder system, and monitoring dashboard. These features can be further refined based on user feedback. This approach of focusing on customer satisfaction, transparency and quality control makes the Agile Methodology an ideal paradigm for this system.



Figure 25 Agile Methodology (Aguayo Publicidad, 2018)

5.2 Requirement Gathering and Analysis

This part elaborates on the process of gathering data and analyzing it. This process helps develop the proposed system to be ideal. The researcher in this process have to gather data in two ways:

primary and secondary data gathering. Primary data gathering refers to findings gathered first hand, directly from users. This helps finalize the system’s functional requirements. Similarly, secondary data refers to when the researcher gathers information from previously existing documents, research papers, and online resources. These methods help understand the system requirements better.

5.2.1. Software Requirements

Component	Requirement
IDE / Development Tool	Android Studio / Visual Studio Code (with Flutter & Dart plugins)
Programming Language	Dart
Framework / SDK	Flutter SDK (latest stable version)
Version Control	(GitHub
Backend Services	Firebase (Authentication, Firestore , Cloud Functions, Storage)
API Integration	REST APIs / JSON / Third-party APIs (e.g., Claude API if AI features used)
Testing Tools	Flutter Test / Integration Test
Package Manager	Pub (pub.dev)
Emulator / Simulator	Android Emulator / iOS Simulator
Debugging Tools	Flutter DevTools / Android Debug Bridge (ADB)
UI/UX Design Tools	Figma / Adobe XD

Figure 26 Software Requirement

5.2.2. Hardware Requirements

Component	Minimum
Device Type	Android smartphone or iPhone
Processor	Quad-core 1.8Ghz
RAM	2 GB
Storage	100 MB free space
Internet	3G/Mobile Data
Notification Support	Push notification capability

Figure 27 Hardware Requirement

5.3. Requirement Analysis

5.3.1. Functional Requirements

- User registration and login
- Aquarium and fish species management
- Subject-specific AI chatbot
- Reminder and notification system
- Fish compatibility checker
- Activity history and care recommendation
- Database storage
- User profile management

MoSCoW Table

Requirements	MoSCoW
User Registration and Login	Must Have
Aquarium and fish species management	Must Have
Subject-specific AI chatbot	Must Have
Reminder and notification	Must Have
Fish compatibility checker	Must Have
Activity history and care recommendation	Should Have
User Profile Management	Should Have
Advanced AI suggestions	Could Have
Multi-aquarium tracking	Could Have
IoT sensor integration	Won't Have
Image-based fish disease detection	Won't Have

Figure 28 Functional Requirement

5.3.2. Non-Functional Requirements

- Fast system response
- Authentication security
- Data privacy
- Scalable backend
- Offline system support
- Reliable notification delivery

MoSCoW Table

Requirements	MoSCoW
Fast system response	Must Have
Authentication security	Must Have
Data Privacy	Must Have
Scalable Backend	Should Have
Offline System Support	Could Have
Advanced security layers	Could Have
Reliable Notification Delivery	Must Have
Efficient Database Performance	Should Have
Full offline system support	Won't Have

Figure 29 Non-Functional Requirement

5.4. System Design

The detailed process of defining the specific technical implementation, modules, data structures and interfaces of the components determined in the architecture is known as system design. It converts the higher-level design into a collection of technical requirements, such as database structures, API descriptions, and internal program flows.

5.4.1. UseCase Diagram

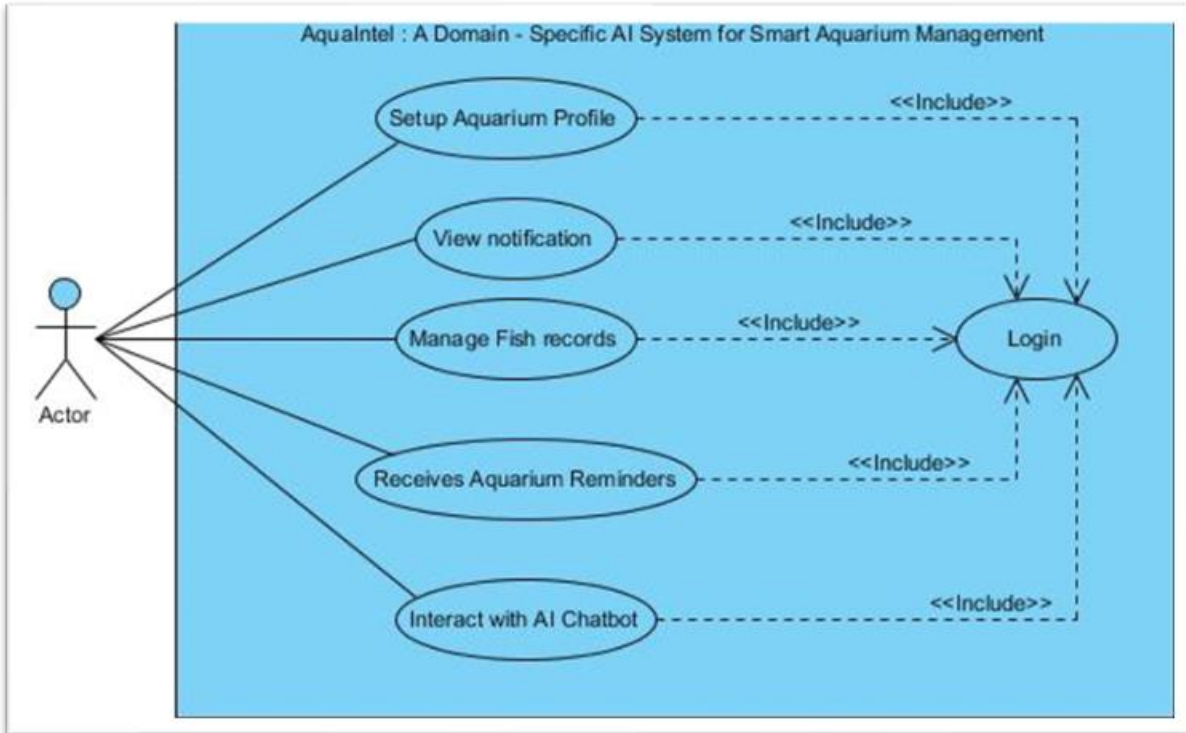


Figure 30 Usecase Diagram of AquaIntel

The AquaIntel Use Case Diagram shows a user-friendly ecosystem in which the authentication is the gatekeeper. Each fundamental interaction, including the first setup of the Aquarium Profiles and Fish Records management to the receipt of Automated Reminders and Viewing Notifications is connected through an obligatory use case with Login by way of an <<include> relationship. This will make sure that any personalized data and tank settings are safe. In addition to its basic utility, the system has a Domain-Restricted AI Chatbot, whereby users can have a specialized conversation about aquatic health and maintenance as long as they are authenticated in the secure mobile environment.

5.4.2. Entity Relationship Diagram

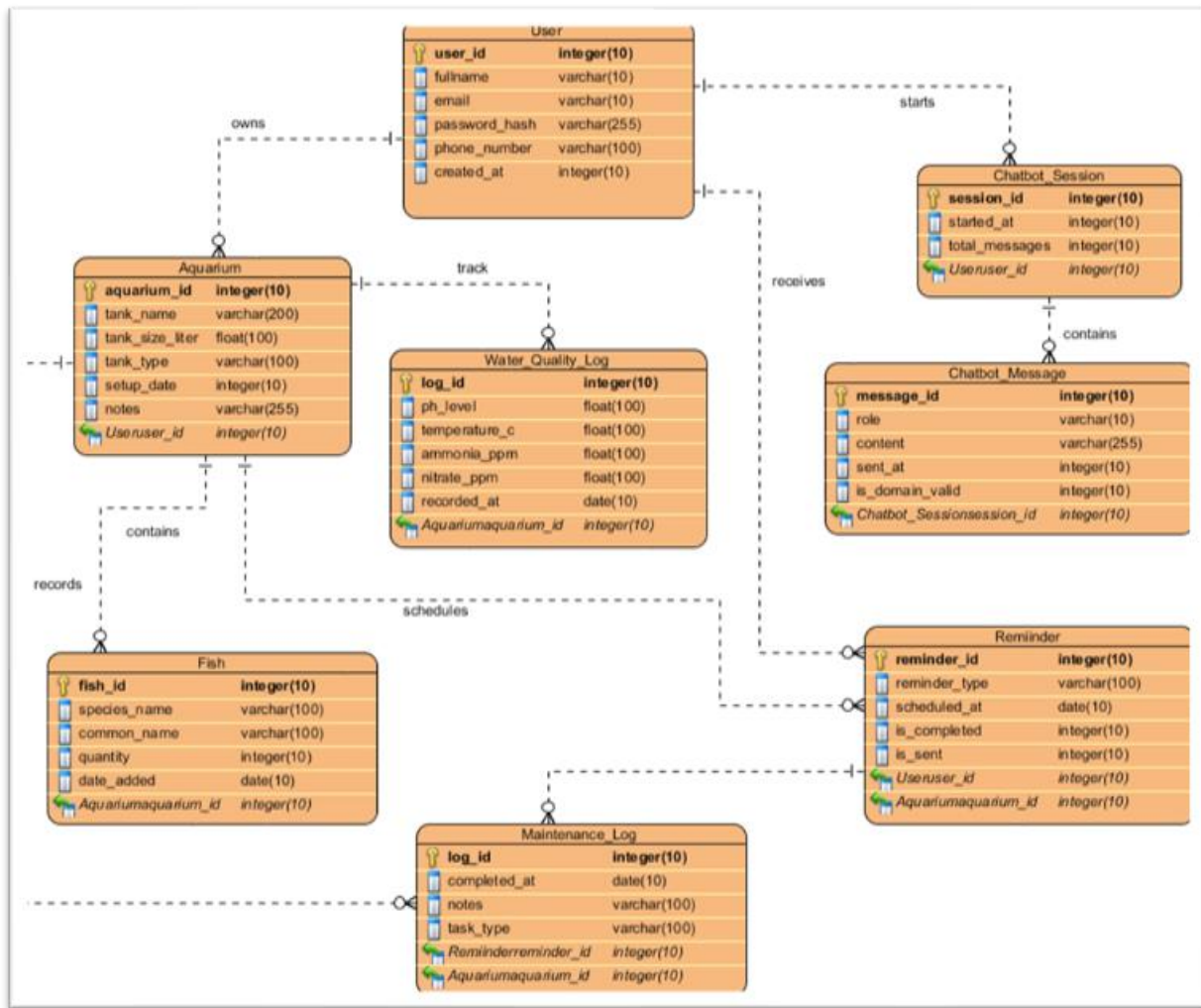


Figure 31 Entity Relationship diagram of AquaIntel

ERD captures 8 entities: USER owns one or more AQUARIUM tanks. Each aquarium contains FISH, generates WATER_QUALITY_LOG entries, and can have multiple REMINDER records. Reminders trigger MAINTENANCE_LOG entries when completed. The chatbot side has CHATBOT_SESSION and CHATBOT_MESSAGE with an is_domain_valid flag to enforce the domain restriction at the data level. The user is the basis of all it; all the tanks, reminders and chats begin with one user profile. The passwordhash value contains a bcrypt hash of the user's password-not the plain text-while createdat tracks the timestamp it was generated. AQUARIUM is the hub entity, positioned centrally due to the fact that nearly all of the other tables are linked to it. This stores the parameters of the physical tank, the sizelitres and typetank, which are used in order to calculate the correct parameters for the indicator.

FISH stores the populations within a tank, not individual fish, species, and the health_status (Healthy/Sick/Dead), which alongside the water parameters is used for the dashboard colour coding. WATERQUALITYLOG is append-only. Every time a user logs a reading a new row is

created. The latest row determines the dashboard health indicator. All rows determine the trend chart. Keeping all the readings (not just the latest) creates the 30-day pH graph.

REMINDER has two booleans that document the life cycle: the is_sent boolean is turned on when the push notification is fired and the is_completed is turned on when the user interacts with it. Completing a reminder automatically creates a MAINTENANCE_LOG entry. CHATBOT_SESSION organises messages within a single conversation. Its total messages counter is rate limited (to regulate the cost of AI API) and can be used to analyze how users interact with the chatbot. domain restriction is stored at the data level in CHATBOT_MESSAGE. The is_domain_valid flag sets all messages true (aquarium queries) and a real AI response; off-topic queries get false and a rejection message. This log can be reviewed to upgrade the filter with time. Maintenance Log MAINTENANCE_LOG This is the history of permanent care of the tank, and is generated automatically when a reminder is filled in. Reminder_id FK is nullable - when a user enters a task manually, which is not in the reminder system, reminder_id is set to null, but the log record remains with what was done and when.

5.4.3. Class Diagram

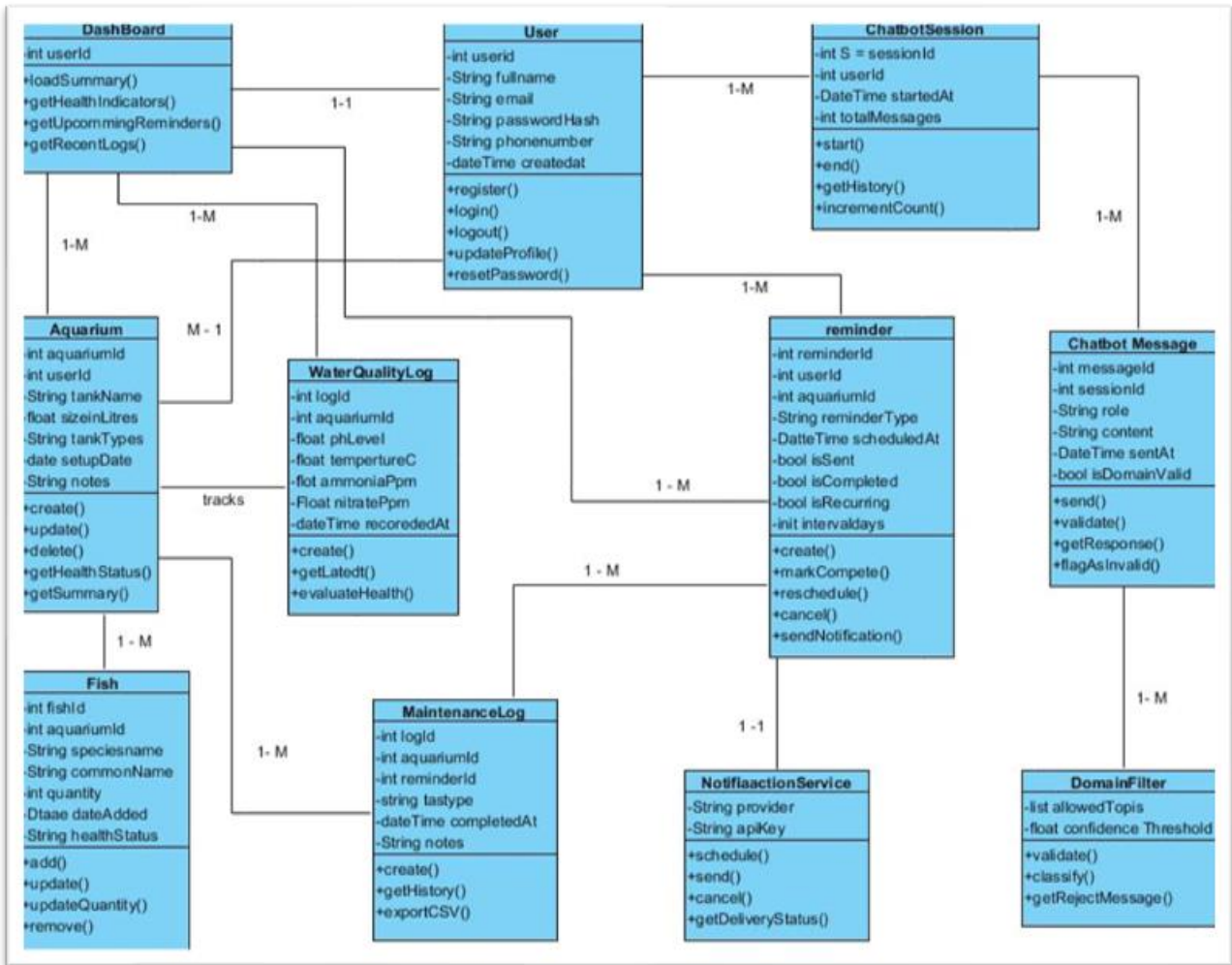


Figure 32 Class Diagram of AquaIntel

System Design Architecture: This is the core architecture of the system. The root entity is a 'User', and a user can have many 'Aquariums'. An aquarium is a hub containing Fish record data, WaterQualityLog data (which aggregates health trends) and a 'Reminder' system which automatically converts finished tasks into 'MaintenanceLogs' for historical storage. The dashboard provides a read only service for the user displaying the various health status and future task deadlines.

The intelligence layer is handled by the ChatbotSessions. Every message must first be filtered by a DomainFilter before it can be presented to the AI. This DomainFilter has stringent aquarium related constraints to assure accuracy and minimize costs. The NotificationService will then handle

any push notifications for maintenance tasks, the internal code uses such methods like `evaluateHealth()` and `validate()` to process user data into actionable domain knowledge.

5.4.4. Activity Diagram

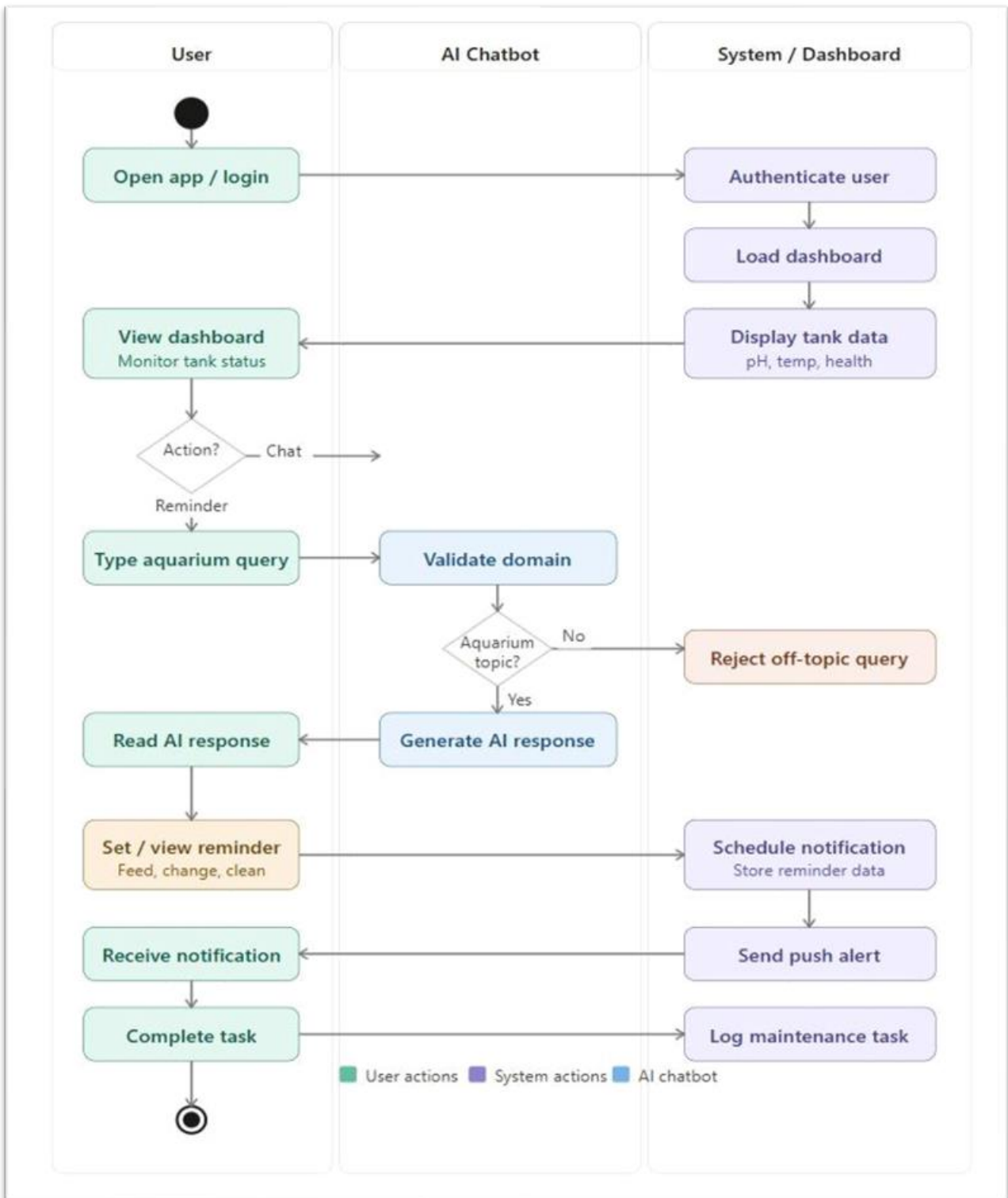


Figure 33 Activity Diagram of AquaIntel

This activity diagram illustrates the interaction of the user with the mobile app, AI chatbot, and the system in the following steps. It starts when the user opens the app and logs in. The system then verifies the user and loads the dashboard. Upon entry, the user is able to see vital tank statistics including pH level, temperature and general health status. The user can choose what to do on the dashboard: either communicate with the chatbot or handle reminders. This section shows the central location of the dashboard where all the aquarium information is shown and controlled.

On the event that the user wants to use the chatbot, he/she types in a query pertaining to the aquarium. The chatbot first checks if the question is within the aquarium domain. In case the query is irrelevant, it is discarded, but in other cases, the AI will produce a response, which the user can read. Conversely, in case the user chooses reminders, then they can create or see things such as feeding, cleaning, or water changes. The system alerts and delivers push notifications when appropriate. Whenever the user does some task, it is documented in the system as a maintenance log. This makes sure that everything taking place is duly monitored and assists in keeping the aquarium in good order.

Activity Diagram shows three kinds of processes on three swim lanes (User, AI Chatbot, System/Dashboard): log-in and authentication flow, real-time dashboard monitoring flow, domain limited interaction between User and AI Chatbot (off-topic conversation denied), reminder/notification loop that ends with maintenance log.

5.4.5. Activity Diagram of Chatbot

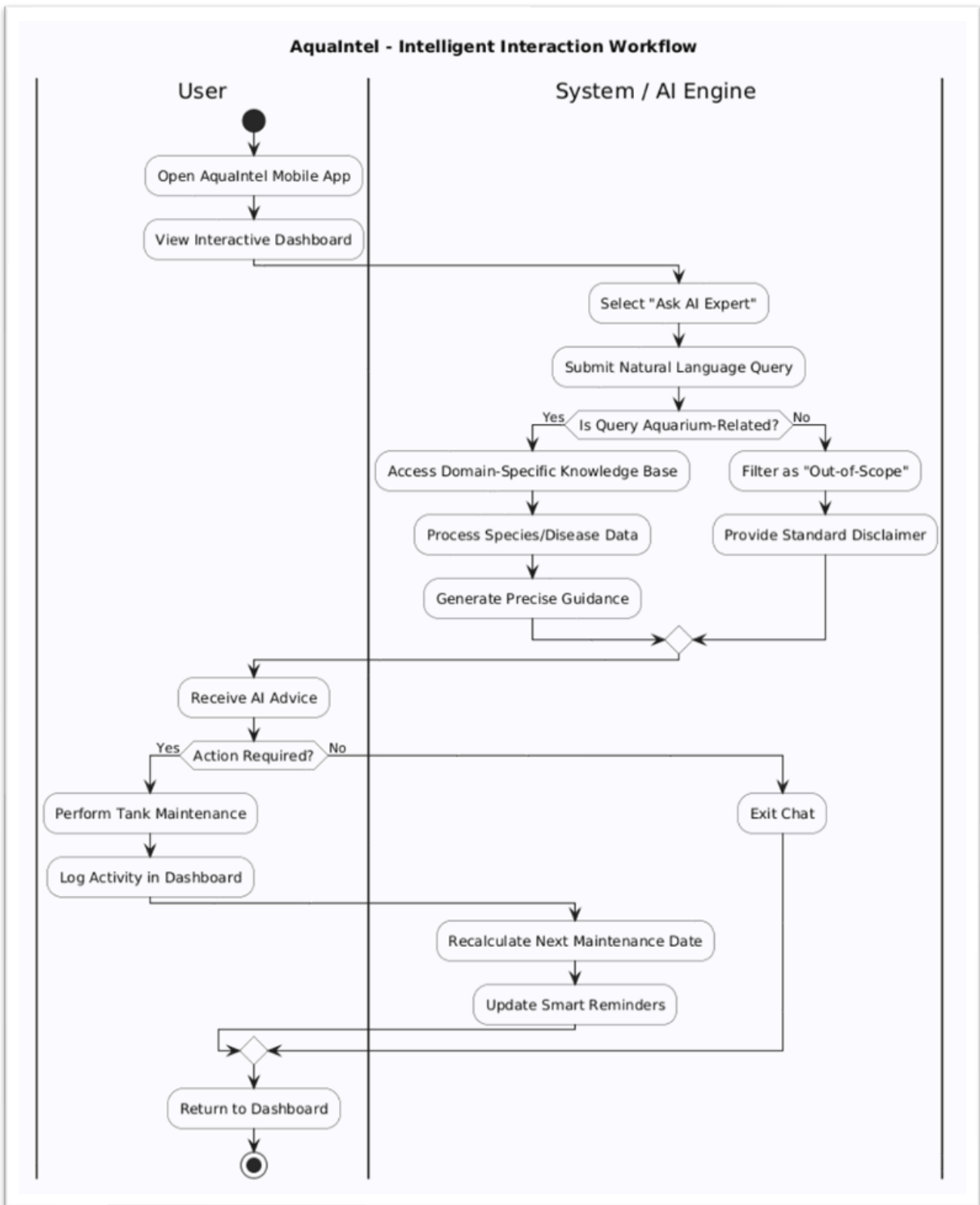


Figure 34 Activity diagram of chatbot

The diagram below describes the process of the Aquaintel AI chatbot and the interaction between the user and the system. It starts when the user opens the Aquaintel mobile app and views the interactive dashboard. At this point, the user clicks the “Ask AI Expert” button and enters a question in natural language. The system then verifies that the query is about aquariums. When the question is pertinent, the AI uses its domain-specific knowledge base, handles information, including fish species or diseases, and provides correct advice. When the query is unrelated, the system filters the query and gives a normal response rather than forwarding the query to the AI engine. The user then makes a decision whether to take any action upon getting the response of the AI. Should there be something to be done, the user can do such actions as maintaining tanks and record the action in the dashboard. The system will subsequently update upcoming schedules by re-calculating the next maintenance date and changing smart reminders. In case no action is necessary the user can just leave the chat. In both instances, the flow is terminated by taking the user back to the dashboard. Such a workflow will make the chatbot relevant in offering advice, decision support, and ensure that the maintenance process in the aquarium is maintained and current.

6. Testing Strategy/Plan

AquaIntel follows a layered testing strategy as it aligns with the Agile methodology. Testing will be conducted incrementally after each sprint to make sure that every component functions correctly before integrating them into the system. The strategy will cover unit testing, integration testing, system testing, and user acceptance testing.

6.1. Testing Objectives

The major objectives of testing include:

- To make sure all functional requirements have been implemented properly
- To validate the subject-specific AI chatbot responds only to questions related to the subject matter
- To confirm maintenance reminders and push notifications are delivered reliably
- To ensure databases store and receive data accurately

6.2. Types of Testing

Unit Testing

Unit testing is applied to each module in the system to make sure that each of those components work as intended. This includes testing the AI chatbot prompt filtering logic, the reminder scheduling functions, and the database query operations.

System Testing

System testing tests the whole application as one. This simulates real-world scenarios. Test cases cover each step the user goes through from registration to chatbot interactions, reminders, etc. Both Android and iOS environments need to be tested to make sure the application is compatible with cross-platform use.

User Acceptance Testing

User Acceptance Testing is conducted with a sample group of aquarium enthusiasts ranging in experience. The participants are asked to operate the app and carry out certain tasks with it. After which the users provide structured feedback. This feedback is then used to evaluate features, and overall satisfaction.

Beta Testing

The users interact with the system in real-time situations for an extended period, enabling the researcher to analyze the operation of the software under actual working conditions. It is during this stage that some problems could arise which have not been detected at any other stage of testing. This is where feedback on the performance of the software is analyzed with emphasis on specific aspects like the AI chatbot, reminder function, and monitoring dashboard.

6.3. Test Cases

Test ID	Feature	Test Description	Expected Result
TC01	User Authentication	Register an account with valid data	Account created and user logs in
TC02	AI Chatbot	Ask an aquarium related question	Accurate response
TC03	AI Chatbot Restriction	Ask an unrelated question	System declines and directs to aquarium topics
TC04	Fish Database	Search for 'Gold Fish' in the database	Correct details shown
TC05	Compatibility Check	Put incompatible fishes to the same tank	Display warning message
TC06	Maintenance Reminder	Set a weekly reminder to change the water	Reminder created and received notification
TC07	Push Notification	Trigger an overdue alert	Push notification to device
TC08	Cross-Platform	Run app on Android and iOS	Consistent functionality on both
TC09	Tank Profile	Create a new tank profile	Tank profile saved

Figure 35 Test Cases

7. Conclusion

The AquaIntel project will offer an intelligent aquarium management mobile app to aquarium hobbyists and shopkeepers alike, both novice and expert. The system is meant to make fish care easier and more convenient with the AI chatbot integration, allowing users to control, diagnose, and give their fish, water state, and tank condition advice, which effectively and reliably works to take care of their fish.

In order to prove the necessity of such a solution, there was a survey with the active participation of 50 respondents, who were representatives of the target user group. The results revealed that both first-time and long-term aquarium owners face real challenges in their management, and the integration of AI chatbots is highly sought not only to provide an overall aquarium management consultancy but also to assist with their diagnosis and guidance on fish health, water chemistry, and tank maintenance.

The design and planning stage of AquaIntel generated some key considerations that will be adopted by the development team:

Simple keyword filtering is not adequate to provide domain restriction in AI systems. The close prompt engineering, context management, and extensive testing will be needed to make sure the chatbot is stably within the aquarium domain and provides the right and valid answers to the users in all cases.

The main survey data, obtained with the help of 50 respondents, evidently demonstrate the challenges that users encounter in their daily aquarium management and creates a solid rationale behind a specialized AI-based mobile app.

A structured methodology of Agile development (with distinct sprints) will be used to offer a structured milestone and to assist in controlling project scope. Particular consideration will be necessary regarding estimating effort required to integrate AI and push notification system since both are known to be more complex to implement compared to standard features.

Flutter has been chosen as the intended cross-platform development platform, as it will be likely to fast-track the delivery of both Android and iOS based on a single codebase. But platform-specific behaviour,

especially in the context of how to deal with push notifications on Android, will need to be taken into account carefully and extensively tested in the implementation stage.

Future Improvement Recommendations:

Although the AquaIntel version 1.0 is scoped to include core features of aquarium management, the following are some of the enhancements that should be considered in future development cycles:

IoT sensor integration: Future releases of AquaIntel may include the ability to connect to IoT sensors that will be able to automatically provide information on water parameters, including pH level, temperature, and ammonia concentration, eliminating the need to manually input the data, and allowing end-users to monitor the data more accurately and easily.

Image-based identification of fish diseases: It would be possible to implement a computer vision model that would enable a user to take a picture of a fish and obtain an automated diagnosis of its health directly in the app, greatly increasing the diagnostic capacity of the system in comparison to the AI chatbot alone.

Increased fish database: Fish database may be expanded not only to common species but also to the rare, regional and marine species. The ability to integrate with an existing external API like FishBase would also enhance the amount of available data on species and enhance the quality of AI-generated care recommendations.

Community functionality: By introducing a community layer to the AquaIntel, users would have the ability to share tank profiles, share care tips and collaboratively contribute knowledge on aquarium management, which would establish a social aspect that fosters user engagement and retention over time, more than the single user.

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9. Appendix

The survey findings of the AquaIntel through the google form of the application from the active participants. There are different section in the google form for aquarium experience, problem and challenges, AquaIntel Concept Validation, AI and smart System, User Experience Expectations,

AquaIntel: A Domain - Specific AI System for Smart Aquarium Management

Time: 3 Minutes

About this Survey:
We are currently in the **early design and development phase** of AquaIntel, and we want to ensure we are solving the right problems from day one.

Because the app hasn't been built yet, your input is incredibly valuable. We want to know what frustrates you most about current hobbyist tools and what "dream features" would keep you inspired to maintain your tanks. Whether it's an easier way to check fish compatibility or a smarter system for tracking water changes, your voice will help us decide what we build first

amogh.shakya001@gmail.com [Switch account](#)

Not shared

1. Age Group

- ☐ Under 18
- ☐ 18-24
- ☐ 25-34
- ☐ 35+

2. Do you currently own an aquarium?

- ☐ Yes
- ☐ No

[Next](#) [Clear form](#)

Aquarium Experience

3. How experienced are you with aquarium keeping?

- ☐ Beginner
- ☐ Intermediate
- ☐ Advanced

4. How long have you been maintaining an aquarium

- ☐ Less than 6 months
- ☐ 6 months - 1 year
- ☐ 1-3 years
- ☐ 3+years

5. What type of aquarium do you have?

- ☐ Freshwater
- ☐ Saltwater

Problems and Challenges

6. What challenges do you face while managing your aquarium?

- ☐ Fish compatibility issues
- ☐ Disease identification
- ☐ Water quality management
- ☐ Feeding guidance
- ☐ all of the above

7. How do you currently get aquarium-related information?

- ☐ Google search
- ☐ YouTube
- ☐ Social media
- ☐ Experts/pet shops
- ☐ Friends/communities

8. How reliable do you find the information available online?

- ☐ Very reliable
- ☐ Somewhat reliable
- ☐ Not reliable

AI and Smart Systems

9. Have you ever used an AI chatbot (like ChatGPT) for guidance?

- ☐ Yes
- ☐ No

10. Do you trust AI to provide accurate aquarium advice?

- ☐ Yes
- ☐ Maybe
- ☐ No

AqualIntel Concept Validation

11. Would you use a system that gives aquarium-specific AI guidance?

- ☐ Yes
- ☐ Maybe
- ☐ No

12. Which features would you find most useful?

- ☐ Fish compatibility checker
- ☐ AI chatbot for aquarium questions
- ☐ Maintenance reminders
- ☐ Disease diagnosis support
- ☐ Aquarium dashboard
- ☐ All the above

13. Would you prefer a system focused only on aquarium knowledge (not general AI)?

- ☐ Yes
- ☐ No

14. How important is accurate, domain-specific advice to you?

- ☐ Very important
- ☐ Important
- ☐ Not important

User Experience Expectations

14. What type of interface do you prefer?

- ☐ Mobile app
- ☐ Web application
- ☐ Both

15. How important is ease of use in such a system?

- ☐ Very important
- ☐ Important
- ☐ Not important



16. Would you use reminders for aquarium maintenance (water change, feeding, etc.)?

- ☐ Yes
- ☐ Maybe
- ☐ No

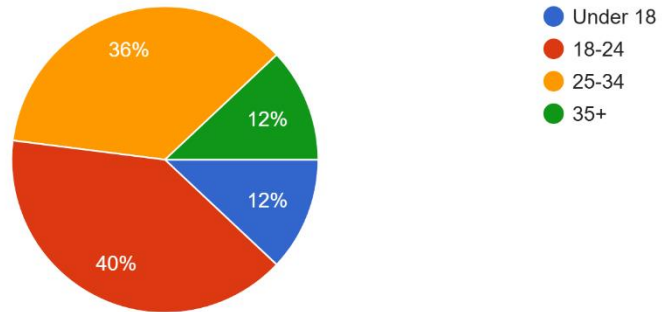
17. Any suggestions for improving Aquaintel?

Your answer

Data Collection of Survey

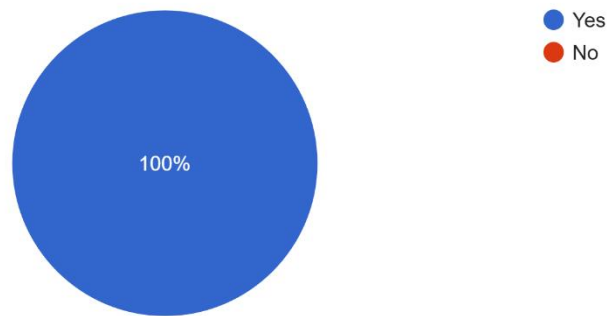
1. Age Group

25 responses



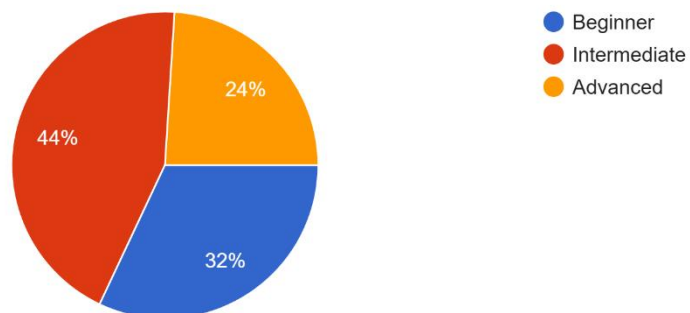
2. Do you currently own an aquarium?

24 responses



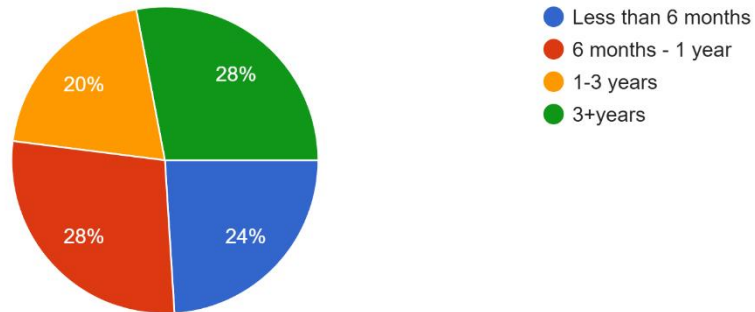
3. How experienced are you with aquarium keeping?

25 responses



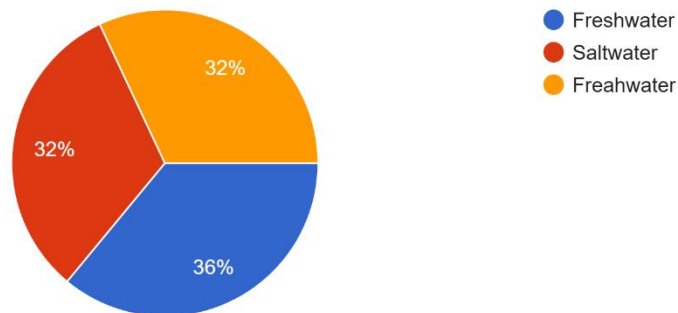
4. How long have you been maintaining an aquarium

25 responses



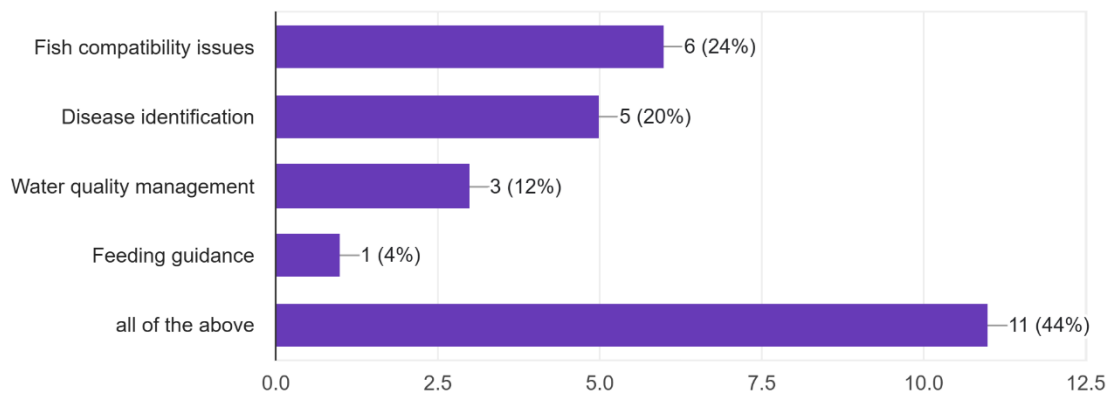
5. What type of aquarium do you have?

25 responses



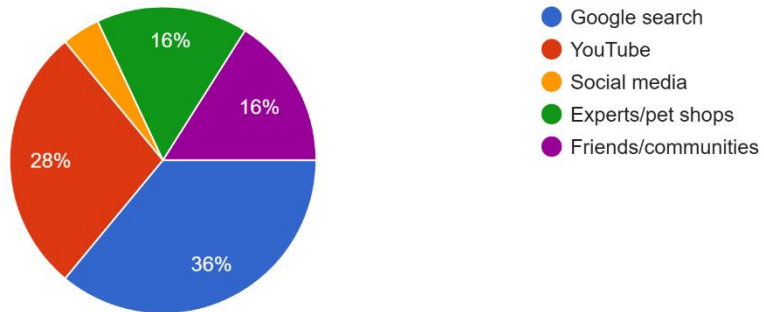
6. What challenges do you face while managing your aquarium?

25 responses



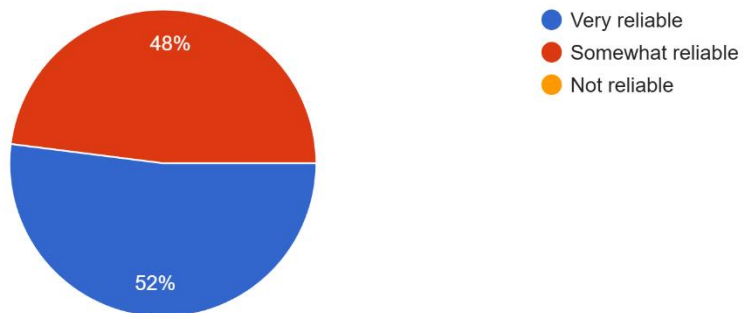
7. How do you currently get aquarium-related information?

25 responses



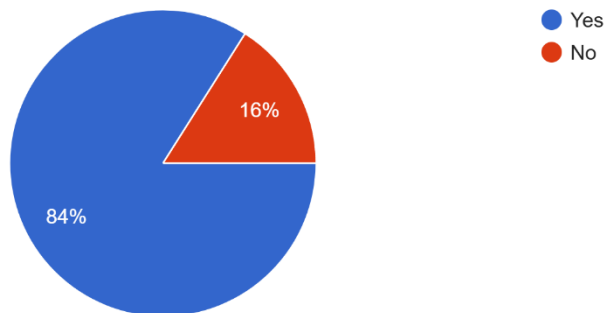
8. How reliable do you find the information available online?

25 responses



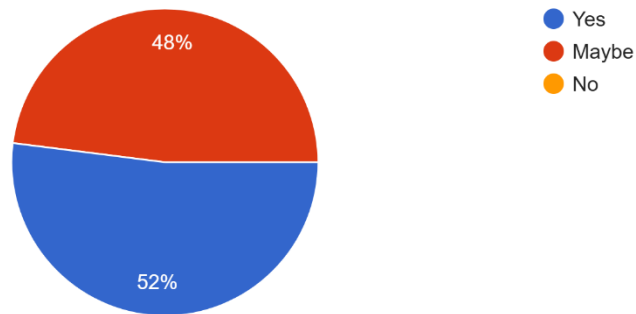
9. Have you ever used an AI chatbot (like ChatGPT) for guidance?

25 responses



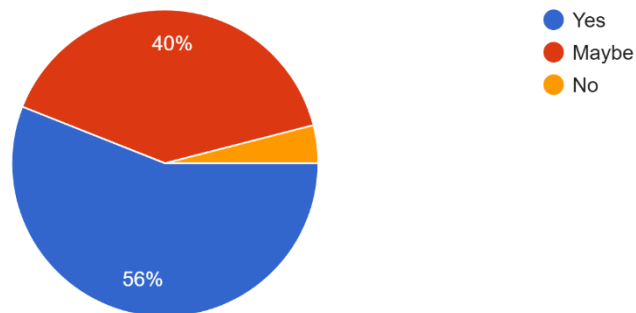
10. Do you trust AI to provide accurate aquarium advice?

25 responses



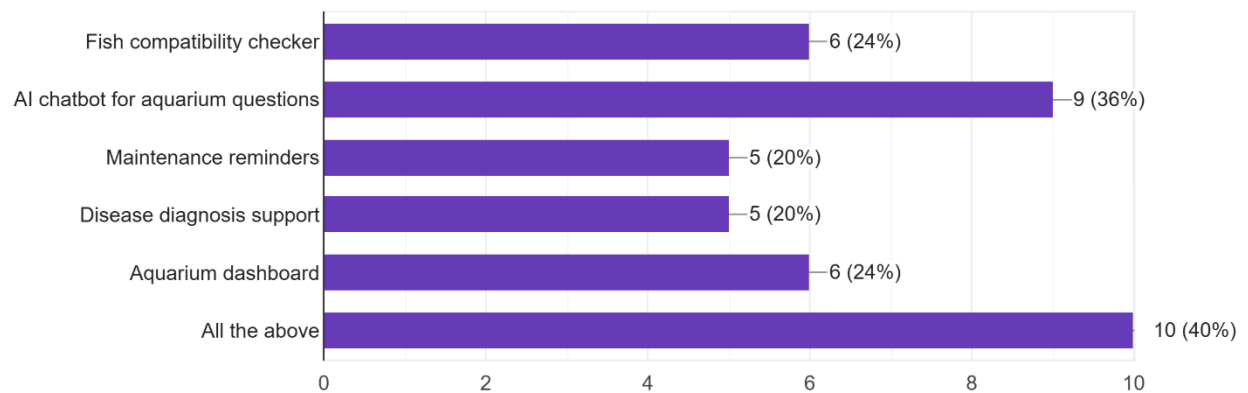
11. Would you use a system that gives aquarium-specific AI guidance?

25 responses



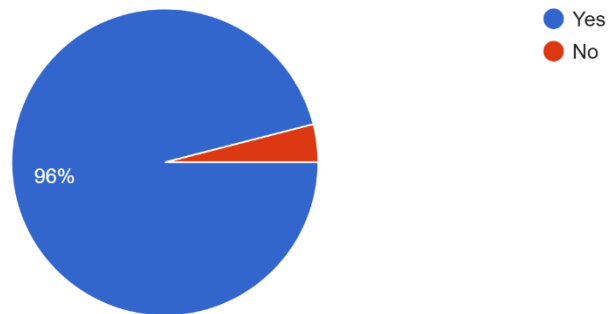
12. Which features would you find most useful?

25 responses



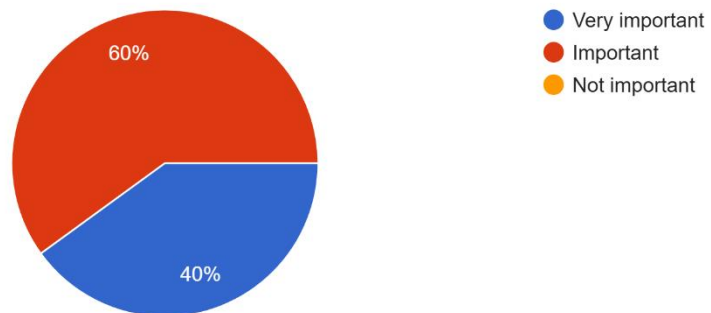
13. Would you prefer a system focused only on aquarium knowledge (not general AI)?

25 responses



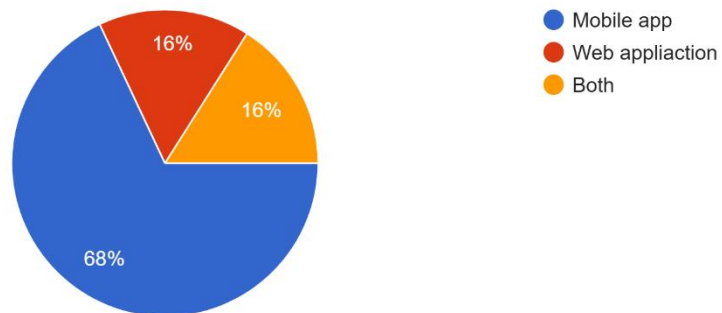
14. How important is accurate, domain-specific advice to you?

25 responses



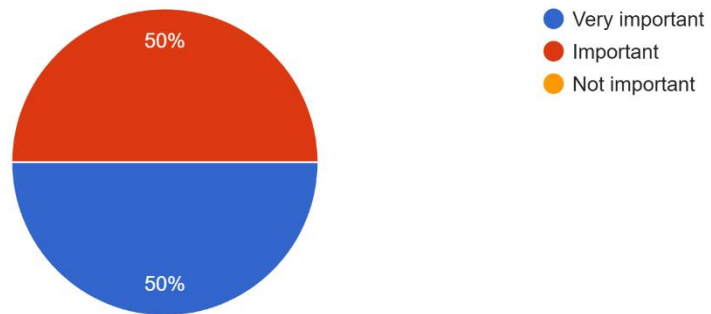
14. What type of interface do you prefer?

25 responses



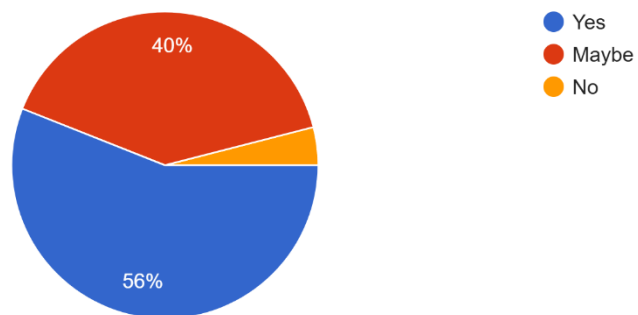
15.How important is ease of use in such a system?

24 responses



16. Would you use reminders for aquarium maintenance (water change, feeding, etc.)?

25 responses



17. Any suggestions for improving AquaIntel?

24 responses

Mobile app should be made

no

No lagging

good UI

Reminder with notifications and alarm

Accuracy

Accuracy & easy access

Mobile app

Accurate result output